1 Logic: random theories, Π_1 truth and Gödel phenomena

Up to constant factors, every version of SAI is the corresponding version of consistent SOI (**??**). This section studies what is specific to data that are sentences of a logic. Random theories can succeed where every r.e. theory fails (Sections 1.1 and 1.2); on Π_1 data the mixture is, up to a constant factor, one random true theory (Section 1.3); Gödel's theorems leave fixed marks on the posterior (Section 1.4); given all its instances, a universal sentence gets belief 1 only when the background and the instances prove it (Section 1.5); and the loss on true arithmetic grows at least polynomially (Section 1.5). Proofs are in Appendix A.

Conventions. $\mathcal{L}_{\mathrm{ar}} = \{0, S, +, \cdot, <\}$ with numerals $\underline{n}$; Δ_0, Σ_1, Π_1 are syntactic classes, and Π_1^* consists of the finite conjunctions of Π_1 sentences and negations of Σ_1 sentences. "Q-equivalent" means provably equivalent in Q; "B is true" means that B has a model whose $\mathcal{L}_{\mathrm{ar}}$-reduct is $\mathbb{N}$; "$B \supseteq \mathsf{Q}$" means $\mathsf{Q} \subseteq \mathrm{Cn}(B)$. $\mathcal{U}$ is the class of Σ_1-unsound deductively closed theories (those proving a false Σ_1 sentence), $\overline{\mathcal{U}}$ its complement, and $P_D(\mathcal{C}) := \mathbb{U}_A(\mathcal{C}; D)/M_A(D)$. (G2) is Gödel's second theorem for r.e. extensions of PA; other standard facts are in Appendix A.1. Losses are in the semimeasure convention unless stated otherwise.

1.1 The lottery

The one-tape loss on a stream is bounded iff $\mathbb{U}(E_\infty) > 0$ (**??**). For SOI on strings, bounded one-tape loss implies that some partial computable labeller extends the stream (Proposition A.3(b)). Does bounded one-tape loss of SAI imply that some consistent r.e. theory proves every datum? It does not; the obstruction is a consistency constraint of unbounded arity.

Let $\mathcal{L}$ contain 0, S and a binary relation symbol Win, and let B be decidable and consistent, not mention Win, and have a model in which the numerals denote pairwise distinct elements (for example $B = \emptyset$, Q or PA). With $N_k := 2^{k+1} - 1$, the *lottery sentences*

$$\mathrm{Lot}_k := \exists x\, \mathrm{Win}(\underline{k}, x) \ \wedge\ \forall x \left(\mathrm{Win}(\underline{k}, x) \to x = \underline{0} \vee \cdots \vee x = \underline{N_k}\right)$$

say that lottery k has a winner among its tickets $0, \ldots, N_k$. The *random lottery theory* G_{lot} reads the tape in consecutive blocks of $k+1$ bits, reads block k as a number $J_k \in \{0, \ldots, N_k\}$, and emits Lot_k, $\mathrm{Win}(\underline{k}, \underline{J_k})$, and $\neg\mathrm{Win}(\underline{k}, \underline{j})$ for $j \le N_k$, $j \neq J_k$. Its theory is consistent with B on every tape.

Theorem 1.1 (The lottery)**.** *There is an infinite stream D_∞ of labelled sentences, computable from $\emptyset'$, such that:*

(a) for every n, $M_A(D_n) \ge 2^{-K(G_{\mathrm{lot}})-c} \prod_{k\ge 0}(1 - 2^{-(k+1)})$; so $\mathbb{U}(E_\infty) > 0$, the cumulative loss is at most $K(G_{\mathrm{lot}}) - \log \prod_{k\ge 0}(1 - 2^{-(k+1)}) + c$, and the posterior converges in total variation to P_∞;

(b) no consistent r.e. theory proves (together with B) every member of $D_\infty^{\pm}$;

(c) $W_A(D_n) \to 0$ and $K_A(D_n) \to \infty$, so the coin-free loss is unbounded; and $K_{\mathrm{fin}}^{\mathcal{L}'}(D_n) \to \infty$ for every language $\mathcal{L}' \supseteq \mathcal{L}$ with computable syntax;

(d) the posterior probability that the tape's program is a deterministic machine (one that emits only under the empty prefix) tends to 0, and no tape in E_∞ has a theory with r.e. deductive closure: the limit posterior lives on random theories whose closure is not r.e.

The product equals $0.288788\ldots$ (a computation; code in the accompanying repository). The stream is a diagonal: if $B \cup \mathsf{W}_e$ is consistent and proves Lot_e, it cannot refute every ticket of

lottery e, and the stream reports as a loser the least ticket j_e^* that it does not refute. So every consistent r.e. theory fails at its own lottery, while a random lottery theory dies at lottery k only if its winner is the reported loser, with probability $2^{-(k+1)}$. Such a stream is neither r.e. nor, for the lottery classes, co-r.e., and the phenomenon needs proof-requiring compatibility (Remark A.2). The random lottery theories share the law $\{\mathrm{Lot}_k\}_k$ and differ only in random contingent facts, the winners: a forced form of the split between laws and random initial conditions (**??**).

Where it cannot happen. The lottery's minimal inconsistent sets $\{\mathrm{Lot}_k\} \cup \{\neg\mathrm{Win}(\underline{k}, \underline{j}) : j \leq N_k\}$ have unbounded size, and at every threshold the majority-vote theory refutes every ticket of the late lotteries, as in Kyburg's lottery paradox (Kyburg, 1961). Bounded arity removes the obstruction: if $\mathbb{U}(E_\infty) > 0$ and the data lie in a decidable sentence class in which every B-inconsistent finite set has a B-inconsistent subset with at most r elements, the majority vote above a Lebesgue density point of E_∞, at a threshold $1 - 1/(r+1)$ that is exact for the argument, is a consistent r.e. theory proving the stream, and the coin-free loss is bounded too (Proposition A.3(a); cf. de Leeuw et al., 1956).

Corollary 1.2 (Bounded loss and r.e. theories)**.** $\mathbb{U}(E_\infty) > 0$ *does not in general imply that a consistent r.e. theory proves every datum of* D_∞ *(Theorem 1.1; inside true arithmetic, Theorem 1.4). It does when the inconsistency arity of the data's sentence class is bounded (Proposition A.3), for streams of* Π_1 *type over a true background* $B \supseteq \mathsf{Q}$ *(Corollary A.14), and for co-r.e. streams over block-decidable sentence classes, which include the lottery classes (Proposition A.5); and if a positive-measure set of true random theories proves the stream, a true r.e. theory does (Proposition A.4).*

We do not claim that these are the only cases. Since consistent SOI agrees with SAI version by version up to constant factors (**????**), the lottery separates one-tape from coin-free consistent SOI, while plain coin-free SOI on sentences has bounded loss on it, through an inconsistent labeller (Corollary A.8).

On the lottery stream the coin-free loss diverges more slowly than any computable unbounded nondecreasing function, and $K_A(D_n) \leq \lceil n/2 \rceil + O(\log n)$ (Corollary A.7, through the proof horizon of Proposition A.6).

Question 1.3. Is there a stream with bounded one-tape loss on which $-\log W_A(D_n) \geq f(n)$ for some computable unbounded nondecreasing f? And does bounded one-tape loss on a co-r.e. stream imply, in general, that a consistent r.e. theory proves it?

Relatives. Random axiom systems drawn by a universal machine are not new (**??** and Appendix A.2); to our knowledge, the comparison of the one-tape law with its own coin-free and Kolmogorov versions, the identification of the succeeding hypotheses as random theories r.e. in the tape, the arity boundary and the arithmetic versions are new.

1.2 Lotteries in arithmetic

Inside arithmetic, self-referential formulas behave like independent atoms with respect to consistency with a true background, with prescribed truth values in $\mathbb{N}$ (Lemma A.9, a variant of the flexible formulas of Kripke, 1962). With them the lottery needs no fresh symbol, and survives even with true data. Let $\langle k, j \rangle$ be a Δ_0 pairing with $k, j \leq \langle k, j \rangle$.

Theorem 1.4 (Lotteries in arithmetic)**.** *Let $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$, and let B be decidable and true with $B \supseteq \mathsf{Q}$. For a formula $\Upsilon(x)$ put $\mathrm{Tick}_{k,j} := \Upsilon(\langle k, j\rangle)$ and $\mathrm{Lot}^{\mathrm{ar}}_k := \bigvee_{j \le N_k} \mathrm{Tick}_{k,j}$. (a) There are a formula Υ_0 and a stream $((\mathrm{Lot}^{\mathrm{ar}}_k, 1), (\mathrm{Tick}_{k,j^*_k}, 0))_{k\ge 0}$ satisfying (a)–(d) of Theorem 1.1 in which every $\mathrm{Tick}_{k,j}$ is Q-equivalent to a false Σ_1 sentence; so the data are Q-equivalent to false Σ_1 sentences labelled true and to true Π^*_1 sentences. (b) There are a formula Υ_1 and a stream of the same form satisfying (a)–(d) of Theorem 1.1 in which every datum is true in $\mathbb{N}$ and Q-equivalent to a Boolean combination of Σ_1 sentences.*

In (b), Υ_1 is chosen by self-reference so that the true winner of lottery k is the least ticket that $B \cup \mathsf{W}_k$ refutes (0 if none); the diagonal reports as the loser a different ticket, one that $B \cup \mathsf{W}_k$ does not refute whenever $B \cup \mathsf{W}_k$ is consistent and proves $\mathrm{Lot}^{\mathrm{ar}}_k$, and that report is true.

Corollary 1.5 (True data, false theories)**.** *On the stream of Theorem 1.4(b): (i) every datum is true, $L_A(D_n)$ is bounded, and $K_A(D_n) - L_A(D_n) \to \infty$; so no coding theorem of the form $K_A \le L_A + f(L_A)$ holds for true data beyond $\Sigma_1 \cup \Pi_1$; (ii) $P_\infty\{\omega : B \cup A_\omega \subseteq \mathrm{Th}(\mathbb{N})\} = 0$: the limit posterior is concentrated on false theories, although every datum is true; (iii) the random lottery theory is Σ_1-unsound on every tape on which, for some k, its winner J_k differs from the true winner of lottery k; conditionally on its survival this has probability* 1*.*

On this stream of true sentences the one-tape inducer predicts with bounded loss and comes to believe, with probability one, false theories: a random theory that guesses winners is right where the data check it and wrong where they do not. The data are adversarial (a Δ_2 diagonal), so this is a possibility result in an idealised setting, not evidence about actual science. Whether also $P_\infty(\overline{\mathcal{U}}) = 0$ is open (Question A.10).

1.3 Π_1 data: derandomisation, a coding theorem, Chaitin's theorem

On Π_1 data positive probability of provability certifies truth. Throughout, $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and B is decidable and true with $B \supseteq \mathsf{Q}$. A data set D is *of* Π_1 *type* if each literal in $D^{\pm}$ is B-equivalent to a Π^*_1 sentence or to a true Σ_1 sentence; then $B \vdash \psi_D \leftrightarrow \pi_D$ for some $\pi_D \in \Pi^*_1$, and $M_A(D) = \mathrm{Bel}_0(\pi_D)$ (Appendix A.4).

Theorem 1.6 (The threshold machine)**.** *Let G_{thr} be the theory machine that, after reading s tape bits u, emits every $\pi \in \Pi^*_1$ for which some lower approximation of $\mathbb{U}(O_\pi)$ exceeds $0.u + 2^{-s}$ ($0.u$ the dyadic rational with digits u). For a tape r that is not a dyadic rational, read as a real, $A_r^{G_{\mathrm{thr}}} = \{\pi \in \Pi^*_1 : \mathbb{U}(O_\pi) > r\}$; this theory is inconsistent if $r < \mathrm{Inc}_0$ and true if $r > \mathrm{Inc}_0$. For every data set D of Π_1 type, $M_A(D) \le \nu_{G_{\mathrm{thr}}}(D) \le 2^{K(G_{\mathrm{thr}})+c} M_A(D)$.*

So on Π_1 data the universal mixture is, up to a constant factor, one random true theory: every Π^*_1 sentence π with $\mathbb{U}(O_\pi)$ above a uniformly random level. The tape helps only by choosing the level, and that costs only what the probability of the datum already charges. Fixing $k+2$ bits of the level just above Inc_0 gives a true r.e. theory T_k of complexity at most $k + K(k) + c$ that proves every Π^*_1 sentence π with $\mathrm{Bel}_0(\pi) \ge 2^{-k}$ (Theorem A.11).

Corollary 1.7 (A coding theorem for Π_1 data)**.** *For every D of Π_1 type, $L_A(D) \le -\log W_A(D) + c \le K_A(D) + c \le L_A(D) + K(\lceil L_A(D) \rceil) + c'$.*

Likewise, if every finite part of a stream is of Π_1 type and $\sup_n L_A(D_n) \le k$, the single true theory $B \cup T_k$ proves every datum (Corollary A.14). For literals no coding theorem holds (**??**). For W_A the bound improves to $-\log W_A \le L_A + K^{\mathrm{pad}}(\lceil L_A \rceil + 1) + c$ (Remark A.12); whether $K_A \le L_A + O(1)$ is open (Question A.13).

Code strings by length-lexicographic rank, so that those of length at most m are the codes at most $\mathrm{rk}_m := 2^{m+1} - 2$, and let $\mathrm{Comp}(p, t, y)$ be Kleene's Δ_0 predicate "t codes the halting computation of V on p, with output y". The Π_1 sentence $\mathrm{Kgt}_m(x) := \forall t \forall p\, (p \leq \underline{\mathrm{rk}_m} \to \neg\mathrm{Comp}(p, t, \underline{x}))$ is true iff $K(x) > m$. If $K(x) > m$, the single true datum $(\mathrm{Kgt}_m(x), 1)$ costs between $m - 2\log m - O(\log\log m)$ and $m + K(m) + c$ bits in the semimeasure and three-valued conventions, and one true theory of complexity at most $m + K(m) + c$ proves all true $\mathrm{Kgt}_m(x)$ at once (Corollary A.15): Chaitin's theorem for random theories. In the normalised convention, with B true and $\mathsf{PA} \subseteq \mathrm{Cn}(B)$ and for every $j \geq 0$, the datum costs at most $\log(1 + 2^{-j})$ for all but at most $2^{2m+K(m)+j+c} + 2^{m+1}$ strings x of each length (Proposition A.16): the m bits are the price of deciding the sentence, not of predicting its truth value.

The boundary of the coding theorem is exact: with $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$, for every $d \geq 2$ there are data D_d with $K_A(D_d) > \log d - 2$ and $L_A(D_d) \leq 2 + c$ made of one datum Q-equivalent to a false Σ_1 sentence and one to a true Π_1^* sentence, and others made of two true data Q-equivalent to Boolean combinations of Σ_1 sentences (Propositions A.18 and A.19); on a stream, the gap is bounded against unbounded (Corollary 1.5). To our knowledge, the threshold just above the unknown Inc_0, the single random threshold theory and the coding theorem for an inductive inference rule are new.

1.4 Gödel phenomena

Proposition 1.8 (The arithmetical shape of the trichotomy)**.** *Let B be decidable and consistent with $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, and let D be consistent. (a) If σ is a true Σ_1 sentence, then $\mathrm{Bel}_D(\sigma) = 1$. (b) If $\pi \in \Pi_1^*$ is false, then $(\mathrm{Bel}_D, \mathrm{Dis}_D, \mathrm{Indep}_D)(\pi) = (0, 1, 0)$; equivalently, $\mathrm{Bel}_D(\pi) > 0$ or $\mathrm{Indep}_D(\pi) > 0$ implies that π is true. (c) If $B \cup D^{\pm}$ is true, $\pi \in \Pi_1^*$ is true and $B \cup D^{\pm} \nvdash \pi$, then $\mathrm{Bel}_D(\pi)$, $\mathrm{Dis}_D(\pi)$, $\mathrm{Indep}_D(\pi)$ are all positive. (d) If $B \cup D^{\pm}$ is true, σ is a false Σ_1 sentence and $B \cup D^{\pm} \nvdash \neg\sigma$, then all three values of σ are positive.*

The inducer errs one-sidedly: on Σ_1 sentences it may believe a falsehood but never disbelieves a truth; on Π_1 sentences (consistency statements, Gödel sentences, $\mathrm{Kgt}_m(x)$) the reverse. So "independent" on a Π_1 question certifies truth, the inducer's form of "the Gödel sentence is true because it is unprovable"; the normalised convention discards exactly this share.

Theorem 1.9 (Soundness is never learned, coin-free)**.** *Let B be decidable and consistent with $\mathsf{PA} \subseteq \mathrm{Cn}(B)$. There is c such that $W_A(\mathcal{U}; D) \geq 2^{-c} W_A(\overline{\mathcal{U}}; D)$ for every consistent D. Hence the coin-free posterior probability of Σ_1-soundness, and a fortiori of truth, is at most $1/(1 + 2^{-c})$ for every D, in particular for every finite part of true arithmetic; and in the Kolmogorov version the shortest compatible Σ_1-unsound theory is at most c bits longer than the shortest compatible Σ_1-sound one.*

Every consistent W_e has a companion $\mathsf{W}_e \cup \{\neg\mathrm{Con}(B \cup \mathsf{W}_e)\}$, consistent by (G2), unsound, and only $O(1)$ bits longer. One-tape it gives $P_D(\mathcal{U}) \geq 2^{L_A(D) - K_A(D) - c}$ (Proposition A.20): confidence in soundness needs a large coding gap. On data of Π_1 type the bound is uniform: one-tape, over a decidable true background containing PA, the posterior probability of Σ_1-soundness is at most $1 - 2^{-c}$, with c independent of the data (Theorem A.21, given (G2)); whether this holds for all data is open (for partial answers see Appendix A.5):

Question 1.10. With B true and $\mathsf{PA} \subseteq \mathrm{Cn}(B)$, is $P_D(\overline{\mathcal{U}})$ bounded away from 1 uniformly over all data D?

On data presenting an r.e. T with $B \cup T$ true and $\mathsf{PA} \subseteq \mathrm{Cn}(B \cup T)$, $\mathrm{Con}(B \cup T)$ keeps positive limit shares on all three answers, and which of "true" and "false" is favoured is a choice of universal machine (Proposition A.22); without truth this fails (for $B = \mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$ and $T = \emptyset$, $\lim \mathrm{Dis}(\mathrm{Con}(B)) = 1$). Checking instances of a decidable property, the classic case of induction in mathematics,[1] confirms nothing when deduction is free: data entailed by B leave the posterior equal to the prior, for M_A, W_A and K_A alike, so with $\mathsf{Q} \subseteq \mathrm{Cn}(B)$ and a Δ_0 formula $P(x)$, consistent data $\{(P(\underline{n}), 1) : n \leq N\}$ leave $\mathrm{Bel}(\forall x\, P(x))$ at its prior value for every N (Proposition A.23): the problem of old evidence and logical omniscience (cf. Hacking, 1967; Garber, 1983). Instances of a fresh predicate do confirm, up to the ω-gap; induction in mathematics needs a reasoner for whom computing $P(\underline{n})$ is costly, such as a logical inductor (Garrabrant et al., 2016a) or a derivation-charged inducer (Appendix A.5).

1.5 The ω-gap

Proposition 1.11 (Anatomy of the ω-gap)**.** *Let $\mathcal{L}$ have a nonempty set* CT *of closed terms, B be decidable and consistent, $\varphi(x)$ have one free variable, and $I := B \cup \{\varphi(t) : t \in \mathrm{CT}\}$ be consistent. Let the stream list $(\varphi(t), 1)$ for all $t \in \mathrm{CT}$, in any order. Then $\mathbb{U}(E_\infty) > 0$, the posterior converges in total variation to the prior restricted to the consistent extensions of I, and the limit trichotomy of $\forall x\, \varphi$ is (i) $(1, 0, 0)$ if $I \vdash \forall x\, \varphi$; (ii) $(0, 1, 0)$ if $I \vdash \neg\forall x\, \varphi$ (I is ω-inconsistent at φ); (iii) otherwise in $(0, 1)^3$, with* $\lim \mathrm{Dis} = P_\infty($*the theory is ω-inconsistent at φ*$)$ *and* $\lim \mathrm{Indep} = P_\infty($*the theory proves every instance and decides $\forall x\, \varphi$ neither way*$)$.

So the Bayesian ω-gap is the ω-incompleteness of the background plus the instances, weighted by prior shares. In particular Bel satisfies the Gaifman condition (Gaifman, 1964) exactly when $I \vdash \forall x\, \varphi$, and otherwise misses it by at least $2^{-K(I)-c}$ (Proposition A.24), the provability analogue of Proposition B.10 of our earlier report (Hänni and Claude (Anthropic), 2026). That condition is a desideratum for credences about truth (Hutter et al., 2013); Bel is a probability of provability, and its failure is not a defect. The universal prior does not close the gap even for true Π_1 sentences with trivially checkable instances: over a decidable true background containing PA, for every k there is such a sentence whose belief stays below $2^{-k+c'}$ after any instance data (Theorem A.25, given (G2)). In arithmetic the gap closes under the ω-rule (with $B \supseteq \mathsf{Q}$) and under full second-order consequence, and in the generative variant only with the universal sentence among the data (Remark A.26). A bound on the number of lines of derivations does not close it either: under a full-support prior the limit belief is then 1 iff a Kreisel-type property holds, and that fails already for a fresh predicate and nine lines (Proposition A.27 and Example A.28).

True arithmetic. On true arithmetic, listed by length in a syntax with short binary numerals, the loss in the semimeasure convention grows at least like n^ξ up to logarithmic terms, for an exponent $\xi > 0$ of the syntax ($\xi \geq 0.0823$ for a stated syntax, by an exact computation; code in the accompanying repository), and every true r.e. theory leaves at least $C'^{-\xi} n^\xi / (2 \log n)$ of the first n sentences undecided for large n, for a constant C' (Theorem A.30 and Corollary A.31); whether the loss is $o(n)$ is open (Question A.33).

[1] The question of how checking instances can confirm a universal mathematical statement first appeared in Hänni's working notes (Hänni, 2026).

A Proofs for random theories, Π_1 data and Gödel phenomena

This appendix proves the results of Section 1 in their order and states and proves the results that the main text only mentions. The notation is that of **??????**.

A.1 Standing facts and conventions

Facts from ??. We use four facts throughout.

- *(Dom)* If $\omega = p\omega'$ with $p \in \operatorname{dom} V$, then $A_\omega = A_{\omega'}^{G_{V(p)}}$ (**??**). So for every theory machine G, every event about the random theory of G is matched, at a cost of a factor $2^{-K(G)}$, by the same event for G_U: the tapes $p\omega'$ with p a shortest program for an index of G carry it. In particular $M_A \geq 2^{-K(\nu)}\nu$ for every theory-machine law ν, and $M_A(D) \geq 2^{-K(e)-c}$ whenever $B \cup \mathsf{W}_e$ is consistent and proves ψ_D (**??**).
- *(Chain)* $2^{-K_A(D)} \leq W_A(D) \leq 2^c M_A(D)$ for consistent D (**??**(c)).
- *(Lim)* If $\mathbb{U}(E_\infty) > 0$, the posterior converges in total variation to $P_\infty = \mathbb{U}(\cdot \mid E_\infty)$ and the cumulative loss is bounded; if $\mathbb{U}(E_\infty) = 0$, the loss tends to ∞. For every Γ with $\mathrm{Cn}(\Gamma) = \mathrm{Cn}(D^{\pm}_\infty)$, complete or not, $E_\infty = \{\omega : B \cup A_\omega \text{ consistent, } \mathrm{Cn}(A_\omega) \supseteq \mathrm{Cn}(\Gamma)\}$ (**??**).
- *(Old)* $M_A(D)$ depends on D only through $\mathrm{Cn}(D^{\pm})$, and data entailed by B change nothing (**??**(c)). The same holds for W_A and K_A, whose definitions mention D only through "proves ψ_D".

$\mathbb{U}(O_\varphi \cap [w])$ is left-c.e. uniformly in φ and w (**??**). For a set $\mathcal{X}$ of tapes and a finite string w we write $\mathbb{U}(\mathcal{X} \mid [w]) = 2^{|w|}\mathbb{U}(\mathcal{X} \cap [w])$. For a class $\mathcal{C}$ of deductively closed theories, $P_D(\mathcal{C}) := \mathbb{U}_A(\mathcal{C}; D)/M_A(D)$ is the posterior probability that the random theory lies in $\mathcal{C}$, and $W_A(\mathcal{C}; D) := \sum_e \mathbf{m}(e)\mathbf{1}[B \cup \mathsf{W}_e \text{ consistent}, \vdash \psi_D, \ \mathrm{Cn}(\mathsf{W}_e) \in \mathcal{C}]$.

Arithmetic. $\mathcal{L}_{\mathrm{ar}} = \{0, S, +, \cdot, <\}$ with numerals $\underline{n} = S^n 0$. The classes Δ_0, Σ_1, Π_1 are syntactic; Π_1^* is the class of finite conjunctions of Π_1 sentences and negations of Σ_1 sentences; a sentence that is equivalent to a member of a class only provably in Q is called Q-*equivalent* to one. "B is true" means that B has a model whose $\mathcal{L}_{\mathrm{ar}}$-reduct is $\mathbb{N}$ (for $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$: $B \subseteq \mathrm{Th}(\mathbb{N})$), likewise "$B \cup T$ is true"; "$B \supseteq \mathsf{Q}$" means $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, likewise for PA. A theory is Σ_1-*sound* if it proves no false Σ_1 sentence. Binary strings are coded by numbers through their length-lexicographic rank (the empty string is 0, then $0, 1, 00, 01, \ldots$); the coding is injective across lengths, the length of a string is computable from its code, and the strings of length at most m are exactly the codes at most $\mathrm{rk}_m := 2^{m+1} - 2$. We use the following standard facts (see Smoryński, 1977; Boolos et al., 2007).

- *(Σ_1 C)* Q proves every true Σ_1 sentence (in particular every true Δ_0 sentence) and refutes every false Δ_0 sentence.
- *(Π_1 T)* Hence, if $B \supseteq \mathsf{Q}$ and $B \cup A$ is consistent, every Π_1^* sentence that $B \cup A$ proves is true: if a conjunct were false, its negation would be a true Σ_1 sentence, provable in Q.
- *(O)* $\mathsf{Q} \vdash \forall z\,(z \leq \underline{n} \to z = \underline{0} \vee \cdots \vee z = \underline{n})$ for every n; so Q unfolds bounded quantifiers with a numeral bound into finite conjunctions and disjunctions.
- *(G2)* For every consistent r.e. presentation Z with $\mathsf{PA} \subseteq \mathrm{Cn}(Z)$, $Z \nvdash \mathrm{Con}(Z)$, where $\mathrm{Con}(Z)$ is the canonical consistency statement of the presentation, a Π_1 sentence.

- *(Diag)* For every formula $\vartheta(z,x)$ there is a formula $\Upsilon(x)$ with $\mathsf{Q} \vdash \forall x\,(\Upsilon(x) \leftrightarrow \vartheta(\ulcorner \Upsilon \urcorner, x))$.
- *(Uniq)* Let $\mathrm{Comp}(p,t,y)$ be Kleene's Δ_0 predicate "t codes the halting computation of V on the program p, with output y". Each halting computation has a unique code, and $\mathsf{PA} \vdash \forall p,t,t',y,y'\,(\mathrm{Comp}(p,t,y) \wedge \mathrm{Comp}(p,t',y') \to t = t' \wedge y = y')$.

Loss conventions. As **??** requires, every loss statement in Section 1 names its convention. Unless stated otherwise losses are in the semimeasure convention: the cumulative loss of D_n is $L_A(D_n) - L_A(\emptyset)$, and $-\log W_A(D_n) + \log W_A(\emptyset)$ for the coin-free version. The lower bounds Theorem 1.1(c), Corollary A.7, Corollary A.15 and Theorem A.30 are statements in this convention. Upper bounds transfer to the normalised convention, since each normalised conditional is at least the raw one; lower bounds do not, and Proposition A.16 shows that the lower bound of Corollary A.15 fails in the normalised convention for most strings. In the three-valued convention the conditional probability of the label 1 on a query φ after two-valued data D is $\mathrm{Bel}_D(\varphi)$, so a lower bound on $-\log \mathrm{Bel}_D(\varphi)$ is a lower bound on the three-valued cost of a true label. Statements about K_A, K_{fin} and the posterior (P_∞, Bel, Dis, Indep) depend on no convention.

A.2 The lottery

Lemma A.1 (Lebesgue density on Cantor space)**.** *Let $\mathcal{X}$ be a Borel set of tapes with $\mathbb{U}(\mathcal{X}) > 0$ and let $\varepsilon \in (0,1)$. Then there is a finite string w with $\mathbb{U}(\mathcal{X} \mid [w]) > 1 - \varepsilon$.*

Proof. By outer regularity there is an open $O \supseteq \mathcal{X}$ with $\mathbb{U}(O) < \mathbb{U}(\mathcal{X})/(1-\varepsilon)$. Write $O = \bigsqcup_i [w_i]$ with $\{w_i\}$ prefix-free. If $\mathbb{U}(\mathcal{X} \cap [w_i]) \le (1-\varepsilon)2^{-|w_i|}$ for every i, summing gives $\mathbb{U}(\mathcal{X}) \le (1-\varepsilon)\mathbb{U}(O) < \mathbb{U}(\mathcal{X})$, a contradiction. □

Proof of Theorem 1.1. The random lottery theory is consistent. Take a model $\mathfrak{M}$ of B in which the numerals denote pairwise distinct elements, and for a tape ω put $\mathrm{Win}^{\mathfrak{M}} := \{(\underline{k}^{\mathfrak{M}}, \underline{J_k}^{\mathfrak{M}}) : k \in \mathbb{N}\}$. Because the numerals are distinct, the only Win-partner of $\underline{k}^{\mathfrak{M}}$ is $\underline{J_k}^{\mathfrak{M}}$, so every sentence G_{lot} emits holds; B does not mention Win. So $B \cup A_\omega^{G_{\mathrm{lot}}}$ is consistent for every ω.

The stream. Let $\mathsf{W}_0, \mathsf{W}_1, \ldots$ enumerate the r.e. sets of $\mathcal{L}$-sentences. For each e define $j_e^* \in \{0, \ldots, N_e\}$: if $B \cup \mathsf{W}_e$ is consistent and proves Lot_e, let j_e^* be the least $j \le N_e$ with $B \cup \mathsf{W}_e \nvdash \neg\mathrm{Win}(\underline{e}, \underline{j})$; otherwise $j_e^* := 0$. In the first case such a j exists: if $B \cup \mathsf{W}_e$ proved $\neg\mathrm{Win}(\underline{e}, \underline{j})$ for all $j \le N_e$, then with the second conjunct of Lot_e and the equality axioms it would prove $\forall x\, \neg\mathrm{Win}(\underline{e}, x)$, contradicting the first conjunct and its consistency. Put

$$D_\infty := (\mathrm{Lot}_0, 1),\ (\mathrm{Win}(\underline{0}, \underline{j_0^*}), 0),\ (\mathrm{Lot}_1, 1),\ (\mathrm{Win}(\underline{1}, \underline{j_1^*}), 0),\ \ldots$$

"$B \cup \mathsf{W}_e$ is consistent" is Π_1, and "$B \cup \mathsf{W}_e \vdash \mathrm{Lot}_e$" and "$B \cup \mathsf{W}_e \vdash \neg\mathrm{Win}(\underline{e}, \underline{j})$" are Σ_1, uniformly; so $e \mapsto j_e^*$ and the stream are computable from $\emptyset'$.

(a) Let $\mathrm{Surv} := \{\omega : J_k(\omega) \neq j_k^* \text{ for all } k\}$. The J_k are independent and uniform on $\{0, \ldots, N_k\}$, so by continuity from above $\mathbb{U}(\mathrm{Surv}) = \prod_k (1 - 2^{-(k+1)})$, which is positive because $\sum_k 2^{-(k+1)} < \infty$. For $\omega \in \mathrm{Surv}$ the theory $B \cup A_\omega^{G_{\mathrm{lot}}}$ is consistent, emits every Lot_k, and emits $\neg\mathrm{Win}(\underline{k}, \underline{j_k^*})$ because $j_k^* \le N_k$ and $j_k^* \neq J_k$. So $\nu_{G_{\mathrm{lot}}}(D_n) \ge \mathbb{U}(\mathrm{Surv})$ for all n, and (Dom) gives the bound. The rest is (Lim), with $M_A(\emptyset) \le 1$ for the bound on the cumulative loss.

(b) If $B \cup \mathsf{W}_e$ is consistent and proves every member of $D_\infty^{\pm}$, it proves Lot_e, so j_e^* was chosen with $B \cup \mathsf{W}_e \nvdash \neg\mathrm{Win}(\underline{e}, \underline{j_e^*})$; but that sentence is in $D_\infty^{\pm}$.

(c) $W_A(D_n) = \sum_e \mathbf{m}(e)\mathbf{1}[B \cup \mathsf{W}_e \text{ consistent}, \vdash \psi_{D_n}]$. Each indicator is non-increasing in n and, by (b), eventually 0; dominated convergence ($\sum_e \mathbf{m}(e) \le 1$) gives $W_A(D_n) \to 0$, and

$K_A \geq -\log W_A$ by (Chain). For a single sentence ψ of $\mathcal{L}'$ with $B \cup \{\psi\}$ consistent, its $\mathcal{L}$-content $\mathrm{Cont}(\psi) := \{\varphi \in \mathrm{Sent}_{\mathcal{L}} : B \cup \{\psi\} \vdash \varphi\}$ is r.e., because $\mathcal{L}'$ has computable syntax, and $B \cup \mathrm{Cont}(\psi)$ is consistent; so by (b) $\mathrm{Cont}(\psi) \not\supseteq D^{\pm}_{\infty}$, and every consistent ψ fails some datum. Only finitely many ψ have $K(\psi) \leq j$, so $K^{\mathcal{L}'}_{\mathrm{fin}}(D_n) \to \infty$.

(d) Let P_{det} be the set of $p \in \mathrm{dom}\, V$ with $G_{V(p)} \subseteq \{\epsilon\} \times \mathrm{Sent}$, and $A^{(p)}$ the set that $G_{V(p)}$ emits. Then

$$\mathbb{U}\{\omega \in E_{D_n} : \omega \text{ begins with some } p \in P_{\mathrm{det}}\} = \sum_{p \in P_{\mathrm{det}}} 2^{-|p|} \mathbf{1}[B \cup A^{(p)} \text{ consistent}, \vdash \psi_{D_n}] \to 0$$

by (b) and dominated convergence; divide by $M_A(D_n)$, which is bounded below by (a). If $\omega \in E_{\infty}$ and $\mathrm{Cn}(B \cup A_{\omega})$ were r.e., it would be a consistent r.e. theory proving every member of $D^{\pm}_{\infty}$, contrary to (b). □

A computation (code in the accompanying repository) confirms the survival probabilities $\prod_{k<K}(1 - 2^{-(k+1)})$ for $K \leq 5$ by enumerating winner vectors, and runs the diagonal against random lists of theories and against an adversarial list in which each listed theory survives until its own lottery.

Remark A.2 (Which streams, which variant). (i) A stream with $\mathbb{U}(E_{\infty}) > 0$ that no consistent r.e. theory proves is not r.e.: otherwise $D^{\pm}_{\infty}$ itself would be such a theory. For the lottery classes it is not co-r.e. either (Proposition A.5). The streams constructed here are Δ_2; an arbitrary stream with these properties need not be arithmetical (for instance, a diagonal against a non-arithmetical list). (ii) The phenomenon needs proof-requiring compatibility: the r.e. theory $\{\mathrm{Lot}_k : k \in \mathbb{N}\}$ is consistent with every finite part of D_{∞}, so under the consistency-requiring variant it stays compatible forever. Proof-requiring compatibility demands that each reported loser be *derived*. (iii) On the stream of Theorem 1.1 every random *finite* theory, in any extended language with computable syntax, fails, because its content is r.e.; so random r.e. theories strictly beat random finite theories there.

Relatives of the lottery. The closest relative in computability theory is that randomness helps for mass problems: a class of positive measure can have no computable member. For a computable bound q, the q-bounded diagonally non-computable functions ($f(n) < q(n)$, and $f(n)$ differs from the value at n of a fixed partial computable diagonal function) form a Π^0_1 class with no computable member; under the uniform product measure on $\prod_n \{0, \ldots, q(n) - 1\}$ it has measure at least $\prod_n (1 - 1/q(n))$, which is positive exactly when $\sum_n 1/q(n) < \infty$. Every Martin-Löf random sequence computes a diagonally non-computable function (Kučera, 1985; see Bienvenu and Patey, 2017), and Bienvenu and Porter (2016) show that the bounded classes are not tt-negligible when $\sum_n 1/q(n) < \infty$ and are deep when the sum diverges. The lottery's survival condition $\sum_k 1/(N_k + 1) = \sum_k 2^{-(k+1)} < \infty$ is this convergence condition, and the diagonal ticket j^*_k plays the role of the diagonal value. The differences: the lottery's diagonal is computed from $\emptyset'$, against r.e. theories rather than partial computable functions; what fails are consistent r.e. theories proving the stream, not computable members of a class; and the class of surviving tapes is the inducer's compatibility event, so its measure is the inducer's likelihood. The modified Demski prior (Garrabrant, 2015; Garrabrant et al., 2016b) also draws random axiom systems from a universal machine, and, for the process as described in those works, it too gives the lottery stream positive probability: with probability at least $2^{-K(G_{\mathrm{lot}})-c} \prod_k (1 - 2^{-(k+1)})$ its first draw is a surviving lottery theory, every later draw is kept only if consistent with it, and the final theory extends it. Its completed theories containing the stream are not r.e., but completions of

extensions of Q never are, so the contrast between random and r.e. hypotheses is not visible in that prior.

Proposition A.3 (Majority vote)**.** *(a) Let $Z \subseteq \mathrm{Sent}$ be decidable and $r \geq 1$, and suppose that every B-inconsistent finite subset of Z contains a B-inconsistent subset with at most r elements. If $D^{\pm}_{\infty} \subseteq Z$ and $\mathbb{U}(E_{\infty}) > 0$, then some consistent r.e. $T \subseteq Z$ contains $D^{\pm}_{\infty}$; hence $\inf_n W_A(D_n) > 0$, and the coin-free loss is bounded too. (b) For the one-tape law M_F of SOI with inputs X and labels Y: if $D_{\infty} : X \rightharpoonup Y$ is infinite and $\lim_n M_F(D_n) > 0$, some partial computable f extends D_{∞}, so the coin-free version W also has bounded loss on D_{∞}.*

Proof. (a) The argument is the majority vote of **??**, with a threshold adapted to the arity. Put $\varepsilon := 1/(r+1)$ and choose w by Lemma A.1 with $\mathbb{U}(E_{\infty} \mid [w]) > 1 - \varepsilon$. Let $T := \{\varphi \in Z : \mathbb{U}(O_{\varphi} \mid [w]) > 1 - \varepsilon\}$; it is r.e., because $\mathbb{U}(O_{\varphi} \mid [w])$ is left-c.e. uniformly in φ. For $\varphi \in D^{\pm}_{\infty}$ we have $O_{\varphi} \supseteq E_{\infty}$, so $\varphi \in T$. Suppose $B \cup T$ is inconsistent. By compactness and the hypothesis, some $\varphi_1, \ldots, \varphi_j \in T$ with $j \leq r$ are jointly B-inconsistent. Then $\mathbb{U}(\bigcap_i O_{\varphi_i} \mid [w]) > 1 - j\varepsilon \geq 1 - r\varepsilon = \varepsilon$, and every tape in that intersection has an inconsistent theory, so $\mathbb{U}(O_{\perp} \mid [w]) > \varepsilon$. But $O_{\perp} \cap E_{\infty} = \emptyset$ gives $\mathbb{U}(O_{\perp} \mid [w]) < \varepsilon$, a contradiction. So T is consistent, and $\inf_n W_A(D_n) \geq \mathbf{m}(e) > 0$ for an index e of T.

(b) $\mathbb{U}\{\omega : f_{\omega} \supseteq D_{\infty}\} = \lim_n M_F(D_n) > 0$ by continuity from above. Choose w with $\mathbb{U}(\{\omega : f_{\omega} \supseteq D_{\infty}\} \mid [w]) > \frac{1}{2}$ and let $f(x) := y$ iff $\mathbb{U}(\{\omega : f_{\omega}(x) = y\} \mid [w]) > \frac{1}{2}$. The events $\{f_{\omega}(x) = y\}$ are effectively open and disjoint in y, so f is a well-defined partial computable function; for $x \in \operatorname{dom} D_{\infty}$ the event $\{f_{\omega}(x) = D_{\infty}(x)\}$ contains $\{f_{\omega} \supseteq D_{\infty}\}$, so $f(x) = D_{\infty}(x)$.

The threshold is exact for the argument. Let $\{\varphi_1, \ldots, \varphi_s\}$ be a minimal B-inconsistent set, and consider a law with inconsistent mass a and mass $(1-a)/s$ on each of s consistent types, type i proving exactly the φ_j with $j \neq i$ (and their consequences). Then every φ_j is proved with probability $a + (s-1)(1-a)/s$, so the majority set at threshold $1-\varepsilon$ contains all of $\varphi_1, \ldots, \varphi_s$ iff $a > 1 - s\varepsilon$, while the consistent mass exceeds $1-\varepsilon$ iff $a < \varepsilon$. Both hold for some a iff $\varepsilon > 1/(s+1)$. So $1/(r+1)$ is the largest threshold at which the argument is guaranteed to return a consistent set, and a family of minimal inconsistent sets of unbounded size defeats every threshold. A computation (code in the accompanying repository) confirms the critical value for $s \leq 12$, and a linear program over all laws, for each of the 166 nonempty families of minimal inconsistent sets on four sentences, finds the same critical value. □

Instances. Literal data of a fresh predicate ($r = 2$, by **??**). Plain SOI, by (b). For the lottery class the sets $\{\mathrm{Lot}_k\} \cup \{\neg\mathrm{Win}(\underline{k}, \underline{j}) : j \leq N_k\}$ are minimal inconsistent, of size $N_k + 2$, so no r works; and for every threshold $1 - \varepsilon$ the majority set refutes every ticket of each lottery k with $2^{k+1} > 1/\varepsilon$, so it is inconsistent: this is the formal content of Kyburg's lottery paradox.

Proposition A.4 (Majority over true theories)**.** *Let $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$ and $B \subseteq \mathrm{Th}(\mathbb{N})$ decidable. If $\mathcal{T}_{\mathrm{true}} := \{\omega : B \cup A_{\omega} \subseteq \mathrm{Th}(\mathbb{N}),\ B \cup A_{\omega} \vdash D^{\pm}_{\infty}\}$ has positive measure, then some r.e. $T \subseteq \mathrm{Th}(\mathbb{N})$ has $B \cup T \vdash D^{\pm}_{\infty}$.*

Proof. $\mathcal{T}_{\mathrm{true}}$ is Borel: $\{\omega : A_{\omega} \subseteq \mathrm{Th}(\mathbb{N})\} = \bigcap_{\chi \text{ false}} \{\omega : \chi \notin A_{\omega}\}$ is closed, and the proving condition is a countable intersection of open sets. Choose w with $\mathbb{U}(\mathcal{T}_{\mathrm{true}} \mid [w]) > \frac{1}{2}$ by Lemma A.1 and let $T := \{\varphi : \mathbb{U}(O_{\varphi} \mid [w]) > \frac{1}{2}\}$, an r.e. set. It contains $D^{\pm}_{\infty}$, since $O_{\varphi} \supseteq \mathcal{T}_{\mathrm{true}}$ for $\varphi \in D^{\pm}_{\infty}$. For $\varphi \in T$, $\mathbb{U}(O_{\varphi} \cap \mathcal{T}_{\mathrm{true}} \mid [w]) > 0$, so a true theory proves φ and φ is true. Hence $T \subseteq \mathrm{Th}(\mathbb{N})$ and $B \cup T$ proves $D^{\pm}_{\infty}$. □

Truth is a constraint of arity one; this is why the majority argument transfers to it, while Σ_1-soundness, which is not a constraint of arity one, escapes it (Question A.10).

Proposition A.5 (Co-r.e. streams over block-decidable classes)**.** *Let $Z \subseteq \mathrm{Sent}$ be decidable and computably partitioned into finite blocks $Z_0, Z_1, \ldots$ (each given by a canonical index) such that (i) for every finite $X \subseteq Z$, $B \cup X$ is consistent iff $B \cup (X \cap Z_k)$ is consistent for every k, and (ii) consistency of $B \cup X$ for finite $X \subseteq Z_k$ is decidable, uniformly in k. If $D_\infty^\pm \subseteq Z$ is co-r.e. and $B \cup D_\infty^\pm$ is consistent, then some consistent decidable $T \subseteq Z$ contains $D_\infty^\pm$.*

Proof. $Z_k \setminus D_\infty^\pm$ is r.e. uniformly in k; let C_k^s be Z_k minus what has been enumerated into it by stage s. As Z_k is finite, $C_k^s = D_\infty^\pm \cap Z_k$ for large s, and that set is consistent with B; so by (ii) the least s_k with $B \cup C_k^{s_k}$ consistent exists and is computable from k. Put $T := \bigcup_k C_k^{s_k}$, a decidable set. $C_k^s \supseteq D_\infty^\pm \cap Z_k$ for all s, so $T \supseteq D_\infty^\pm$, and $B \cup T$ is consistent by compactness and (i). □

Instances. (1) The lottery class $Z = \{\mathrm{Lot}_k\} \cup \{\mathrm{Win}(\underline{k}, \underline{j})^b : j \le N_k,\ b \in \{0,1\}\}$ with blocks by k: under the hypotheses of Theorem 1.1 a finite $X \subseteq Z$ is consistent with B iff no block contains a complementary pair, or Lot_k together with all $N_k + 1$ negative literals of lottery k (choose the Win-row of each $\underline{k}$ separately in a model with distinct numerals); so (i) and (ii) hold. A computation (code in the accompanying repository) checks this criterion against satisfiability in all structures with at most five elements, for $B = \emptyset$ and two blocks. (2) The arithmetic lottery classes of Theorem 1.4: by Lemma A.9(i) every propositionally satisfiable finite set of Boolean combinations of the tickets is consistent with B, so consistency is a propositional, decidable and blockwise property. Hence every stream of these kinds that answers the question of Corollary 1.2 negatively is neither r.e. (Remark A.2) nor co-r.e. Whether some co-r.e. stream over a sentence class on which consistency is not decidable has $\mathbb{U}(E_\infty) > 0$ and no consistent r.e. theory proving it is open (it is the second part of Question 1.3). The adversary in the proof waits, in each block, for a stage at which the remaining data are certifiably consistent; a counterexample must deny it any such certificate at every finite stage.

Proof of Corollary 1.2. Theorem 1.1(a),(b) give the negative answer, and Theorem 1.4 gives one inside true arithmetic. The positive cases are Proposition A.3(a), Corollary A.14 (for a stream of Π_1 type, bounded loss gives $\sup_n L_A(D_n) \le k$ for some k), Proposition A.5 (with $B \cup D_\infty^\pm$ consistent because $\mathbb{U}(E_\infty) > 0$), and Proposition A.4. □

For a theory machine G and a finite string u, A_u^G is the set of axioms emitted under prefixes of u (**??**).

Proposition A.6 (The proof horizon bounds the coin-free loss)**.** *Let G be a theory machine and D_∞ a stream with*

$$a_G := \mathbb{U}\{\omega : B \cup A_\omega^G \text{ consistent},\ B \cup A_\omega^G \vdash D_\infty^\pm\} > 0.$$

Let the proof horizon $h_G(n)$ *be the least m with $\mathbb{U}\{\omega : B \cup A_\omega^G$ consistent, $B \cup A_{\omega_{1:m}}^G \vdash \psi_{D_n}\} \ge a_G/2$. Then, in the semimeasure convention, for every n,*

$$-\log W_A(D_n) \ \le\ K^{\mathrm{pad}}(h_G(n)) + K(G) + \log(2/a_G) + c.$$

Proof. $h_G(n)$ exists: a derivation of ψ_{D_n} uses finitely many axioms, all emitted under some finite prefix, so the events in the definition increase in m to a set containing the one that defines a_G. Fix $m \ge h_G(n)$ and let Good_m be the set of $u \in \{0,1\}^m$ such that $B \cup A_u^G \vdash \psi_{D_n}$ and $[u]$ contains a tape ω with $B \cup A_\omega^G$ consistent. For $u \in \mathrm{Good}_m$, $A_u^G \subseteq A_\omega^G$ for such an ω, so $B \cup A_u^G$ is consistent and proves ψ_{D_n}; and $|\mathrm{Good}_m| 2^{-m} \ge a_G/2$. The set A_u^G is r.e. with an index $e(G,u)$ computable

from (G,u) and injective in u (write u into the index). Hence, using $K(e(G,u)) \le K(G)+K(u)+c$ and $K(u) \le m + K(m) + c$ for $|u| = m$,

$$W_A(D_n) \ge \sum_{u \in \mathrm{Good}_m} \mathbf{m}(e(G,u)) \ge \sum_{u \in \mathrm{Good}_m} 2^{-K(G)-m-K(m)-c} \ge \tfrac{a_G}{2}\, 2^{-K(G)-K(m)-c}.$$

Minimise $K(m)$ over $m \ge h_G(n)$. □

Corollary A.7 (Rate on the lottery stream)**.** *On the stream of Theorem 1.1, in the semimeasure convention:*

(a) $-\log W_A(D_n) \le K^{\mathrm{pad}}(n) + c$ *for all* n*. Hence the coin-free loss diverges, but more slowly than any computable unbounded nondecreasing function: for every such* f*,* $-\log W_A(D_n) \le \frac{2}{3} f(n) + c_f$*.*

(b) $K_A(D_n) \le \lceil n/2 \rceil + O(\log n)$*, and* $K_A(D_n) \to \infty$*.*

(c) Batch lottery. *Give lottery* k *the* 2^{2k+1} *tickets* $0, \dots, 2^{2k+1}-1$ *and diagonalise in it against every* $e < 2^k$ *at once. Then Theorem 1.1(a)–(d) hold with the same product,* $W_A(D_n) \le \sum_{e \ge 2^k} \mathbf{m}(e)$ *once lottery* k *is complete, and again* $-\log W_A(D_n) \le K^{\mathrm{pad}}(n) + c$*.*

The restriction to nondecreasing functions in (a) is needed: a computable unbounded function that vanishes on the odd numbers is not eventually above K^{pad}.

Proof. (a) For $G = G_{\mathrm{lot}}$, every tape gives a consistent theory, and $a_G = \mathbb{U}(\mathrm{Surv}) > 0$. For $\omega \in \mathrm{Surv}$ the axioms needed for D_n are emitted once the first $\lceil n/2 \rceil$ blocks are read, that is, by the prefix of length $m(n) := \sum_{k < \lceil n/2 \rceil} (k+1)$; so $h_{G_{\mathrm{lot}}}(n) \le m(n)$, which is computable from n, and $K^{\mathrm{pad}}(h_{G_{\mathrm{lot}}}(n)) \le K(m(n)) \le K(n) + c$. Proposition A.6 gives $-\log W_A(D_n) \le K(n) + c'$. As $D_n \subseteq D_{n'}$ for $n \le n'$, $W_A(D_n) \ge W_A(D_{n'})$, so $-\log W_A(D_n) \le \min_{n' \ge n} K(n') + c' = K^{\mathrm{pad}}(n) + c'$. *Slow growth.* For a computable nondecreasing unbounded g let $\eta_g(j)$ be the least n' with $g(n') \ge j$, a computable function. Then $\eta_g(g(n)+1) > n$, so $K^{\mathrm{pad}}(n) \le K(\eta_g(g(n)+1)) \le 2\log(g(n)+1) + c$. For a computable unbounded nondecreasing f, apply this to $g := \lfloor 2^{f/3} \rfloor$, which gives $K^{\mathrm{pad}}(n) \le \frac{2}{3} f(n) + c_f$. Divergence is Theorem 1.1(c).

(b) Let $K := \lceil n/2 \rceil$. The tickets j^*_k, $k < K$, are determined by the truth values of the Σ_1 facts "$B \cup \mathsf{W}_k \vdash \mathrm{Lot}_k$" and "$B \cup \mathsf{W}_k \vdash \neg\mathrm{Win}(\underline{k}, \underline{j})$" ($j \le N_k$): j^*_k is the least unrefuted ticket if W_k proves Lot_k and some ticket is unrefuted, and 0 otherwise (an inconsistent W_k refutes every ticket). There are $2^{K+1} + K - 2$ such facts. Given n and the number of true ones ($K+2$ bits, given K), enumerate proofs until that many facts are found true. So $K(\psi_{D_n}) \le K + O(\log n)$, and $K_A(D_n) \le K(\psi_{D_n}) + c$ by **??**(c). $K_A(D_n) \to \infty$ is Theorem 1.1(c).

(c) The data of lottery k are $(\mathrm{Lot}_k, 1)$ and $(\mathrm{Win}(\underline{k}, \underline{j^*_{k,e}}), 0)$ for $e < 2^k$, where Lot_k now names the tickets $0, \dots, 2^{2k+1}-1$ and $j^*_{k,e}$ is the least ticket that $B \cup \mathsf{W}_e$ does not refute if $B \cup \mathsf{W}_e$ is consistent and proves Lot_k, and 0 otherwise. The random theory reads a block of $2k+1$ bits for lottery k. At most 2^k tickets are reported, so it survives lottery k with probability at least $1 - 2^k/2^{2k+1} = 1 - 2^{-(k+1)}$, and the proof of Theorem 1.1 goes through with the same product. After lottery k every $e < 2^k$ has failed, which gives the bound on W_A. At most n lotteries are complete after n data, so the horizon is at most $\sum_{k<n}(2k+1) = n^2$, which is computable from n, and (a)'s argument applies. □

A computation (code in the accompanying repository) checks the bookkeeping of (b) and the survival probabilities of the batch lottery for three lotteries.

Corollary A.8 (The lottery separates the versions of consistent SOI)**.** *On the stream of Theorem 1.1: (a) $M_F^{\mathrm{cons}}(D_n)$ is bounded below, so one-tape consistent SOI has bounded loss; (b) $W^{\mathrm{cons}}(D_n) \to 0$ and $W_A(D_n) \to 0$, so coin-free consistent SOI has unbounded loss; (c) plain coin-free SOI on sentences has bounded loss: $W^{\mathrm{all}}(D_n) \geq \mathbf{m}(f) > 0$ for a partial computable f extending D_∞, and every such f is inconsistent.*

Proof. **??** gives $M_F^{\mathrm{cons}} \asymp M_A$, and **??** gives $W^{\mathrm{cons}} \asymp W_A$. (a) $M_F^{\mathrm{cons}}(D_n) \geq 2^{-c} M_A(D_n)$ is bounded below by Theorem 1.1(a). (b) $W^{\mathrm{cons}}(D_n) \leq 2^c W_A(D_n) \to 0$ by Theorem 1.1(c). (c) $M_F(D_n) \geq M_F^{\mathrm{cons}}(D_n)$ is bounded below, so by Proposition A.3(b), applied with inputs $\mathrm{Sent}_{\mathcal{L}}$ and labels $\{0,1\}$, some partial computable f extends D_∞, and $W^{\mathrm{all}}(D_n) \geq \mathbf{m}(f) > 0$. Such an f is inconsistent: a consistent one would give $W^{\mathrm{cons}}(D_n) \geq \mathbf{m}(f) > 0$, against (b). □

A.3 Lotteries in arithmetic

The lottery needs a family of sentences that behave, with respect to consistency with B, like independent atoms. Inside arithmetic with a true background, self-referential ("flexible") formulas provide them. Lemma A.9 is a self-contained proof of the weak form we need, in a version that also prescribes the truth values in $\mathbb{N}$. It is the standard self-referential construction behind Kripke's flexible formulas (Kripke, 1962) and Mostowski's independent sequences (Mostowski, 1961); Woodin's universal algorithm is a much stronger statement of the same kind (Woodin, 2011; Blanck and Enayat, 2017; Hamkins, 2018). We did not find the version with a prescribed truth set stated in this form, but it is a routine variant, and we claim no novelty for it. For a code z let $\varphi_{[z]}$ be the formula with code z.

Lemma A.9 (Flexible formulas with prescribed truth)**.** *Let $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$ and let B be decidable and true with $B \supseteq \mathsf{Q}$. Write $\mathrm{In}(y,e)$ for a Δ_0 formula saying "y is in the finite set with canonical index e", and for a finite $F \subseteq \mathbb{N}$ with index e_F write $y \in F$ for $\mathrm{In}(y, \underline{e_F})$. For every formula $\Lambda(z,x)$ there is a formula $\Upsilon(x)$ such that:*

- *(i) $B + \forall y\,(\Upsilon(y) \leftrightarrow y \in F)$ is consistent for every finite $F \subseteq \mathbb{N}$; hence every finite pattern $\{\Upsilon(\underline{n})^{b_n} : n \in F'\}$ is consistent with B;*
- *(ii) $\mathbb{N} \models \Upsilon(\underline{n})$ iff $\mathbb{N} \models \Lambda(\ulcorner \Upsilon \urcorner, \underline{n})$, for every n;*
- *(iii) $\mathsf{Q} \vdash \forall x\,(\Upsilon(x) \leftrightarrow \sigma_\Upsilon(x) \vee (\pi_\Upsilon \wedge \Lambda(\ulcorner \Upsilon \urcorner, x)))$ for a Σ_1 formula $\sigma_\Upsilon(x)$ that is false of every number and a true Π_1 sentence π_Υ.*

Proof. Fix a computable enumeration $\tau_0, \tau_1, \ldots$ of the theorems of B. Call a sentence *of form* (z,e) if it is $\neg\forall y\,(\varphi_{[z]}(y) \leftrightarrow \mathrm{In}(y, \underline{e}))$; whether τ_s is of form (z,e) for some e, and for which e, is decidable from (s,z). Let $\sigma_0(z,x)$ be the Σ_1 formula "there are s and e such that τ_s is of form (z,e), no $\tau_{s'}$ with $s' < s$ is of form (z,e') for any e', and $\mathrm{In}(x,e)$" (the computation of $\tau_0, \ldots, \tau_s$ is an existentially quantified witness, checked by a Δ_0 formula), and $\pi_0(z)$ the Π_1 formula "no τ_s is of form (z,e) for any e". By (Diag), applied to $\vartheta(z,x) := \sigma_0(z,x) \vee (\pi_0(z) \wedge \Lambda(z,x))$, take Υ with

$$\mathsf{Q} \vdash \forall x\,(\Upsilon(x) \leftrightarrow \sigma_0(\ulcorner \Upsilon \urcorner, x) \vee (\pi_0(\ulcorner \Upsilon \urcorner) \wedge \Lambda(\ulcorner \Upsilon \urcorner, x))).$$

As Q is true, the same equivalence holds in $\mathbb{N}$. Suppose some τ_s is of form $(\ulcorner \Upsilon \urcorner, e)$; let s_0 be the least such s, with index e_0 and set F_0. In $\mathbb{N}$, $\sigma_0(\ulcorner \Upsilon \urcorner, n)$ holds iff $n \in F_0$, and $\pi_0(\ulcorner \Upsilon \urcorner)$ fails; so $\mathbb{N} \models \forall y\,(\Upsilon(y) \leftrightarrow y \in F_0)$. But τ_{s_0}, the negation of this sentence, is a theorem of the true theory B, a contradiction. Hence no theorem of B is of form $(\ulcorner \Upsilon \urcorner, e)$, which is (i); the pattern follows from $\forall y\,(\Upsilon(y) \leftrightarrow y \in F)$ with $F := \{n \in F' : b_n = 1\}$, since Q decides each $\underline{n} \in F$ correctly by

(Σ_1C). Moreover $\pi_0(\ulcorner\Upsilon\urcorner)$ is true and $\sigma_0(\ulcorner\Upsilon\urcorner, \underline{n})$ is false for every n, which gives (ii), and (iii) with $\sigma_\Upsilon(x) := \sigma_0(\ulcorner\Upsilon\urcorner, x)$ and $\pi_\Upsilon := \pi_0(\ulcorner\Upsilon\urcorner)$. □

Let $\langle k, j\rangle$ be a Δ_0 pairing with $k, j \le \langle k, j\rangle$.

Proof of Theorem 1.4. The tickets. In (a), Υ_0 is Lemma A.9's formula for $\Lambda(z, x) :\equiv (x \neq x)$. In (b), let $\mathrm{Ref}(z, k, j)$ be the Σ_1 formula "$B \cup \mathsf{W}_k \vdash \neg\varphi_{[z]}(\underline{\langle k, j\rangle})$", and let $\Lambda(z, x)$ be the formula, with all quantifiers outside Ref bounded,

$$\exists k \le x\, \exists t \le x \Big(x = \langle k, t\rangle \wedge t \le N_k \\ \wedge \big[(\mathrm{Ref}(z, k, t) \wedge \forall i < t\, \neg\mathrm{Ref}(z, k, i)) \vee (t = 0 \wedge \forall i \le N_k\, \neg\mathrm{Ref}(z, k, i))\big]\Big);$$

Υ_1 is the lemma's formula for this Λ. In both cases $\mathrm{Tick}_{k,j} := \Upsilon(\underline{\langle k, j\rangle})$ for the formula $\Upsilon \in \{\Upsilon_0, \Upsilon_1\}$, and $\mathrm{Lot}^{\mathrm{ar}}_k := \bigvee_{j \le N_k} \mathrm{Tick}_{k,j}$. In (a), j^*_e is defined as in Theorem 1.1, with $\mathrm{Tick}_{e,j}$ in place of $\mathrm{Win}(\underline{e}, \underline{j})$. In (b), let j°_k be the least $j \le N_k$ with $B \cup \mathsf{W}_k \vdash \neg\mathrm{Tick}_{k,j}$ ($j^\circ_k := 0$ if there is none), and let j^*_k be the least $j \le N_k$ with $j \neq j^\circ_k$ and $B \cup \mathsf{W}_k \nvdash \neg\mathrm{Tick}_{k,j}$ if $B \cup \mathsf{W}_k$ is consistent and proves $\mathrm{Lot}^{\mathrm{ar}}_k$, and the least $j \neq j^\circ_k$ otherwise. The stream is $((\mathrm{Lot}^{\mathrm{ar}}_k, 1), (\mathrm{Tick}_{k,j^*_k}, 0))_{k \ge 0}$.

The random theory. $G^{\mathrm{ar}}_{\mathrm{lot}}$ reads the tape in blocks as G_{lot} does and emits $\mathrm{Lot}^{\mathrm{ar}}_k$, Tick_{k,J_k} and $\neg\mathrm{Tick}_{k,j}$ ($j \le N_k$, $j \neq J_k$) after reading block k. A finite part of its output is implied by $B + \forall y\,(\Upsilon(y) \leftrightarrow y \in F)$ with $F := \{\langle k, J_k\rangle : k \le K\}$ for some K, so by Lemma A.9(i) and compactness its theory is consistent with B on every tape.

The diagonal. If $B \cup \mathsf{W}_e$ is consistent and proves $\mathrm{Lot}^{\mathrm{ar}}_e$, it does not refute every $\mathrm{Tick}_{e,j}$, $j \le N_e$, since $\mathrm{Lot}^{\mathrm{ar}}_e \wedge \bigwedge_j \neg\mathrm{Tick}_{e,j}$ is propositionally contradictory. In (b), j^*_e exists in this case: if W_e refutes some ticket, then j°_e is refuted and every unrefuted ticket differs from j°_e; if it refutes none, then $j^\circ_e = 0$ and $j = 1$ is unrefuted ($N_e \ge 1$). Everything else is the proof of Theorem 1.1 verbatim, with Tick for Win: survival probability $\prod_k (1 - 2^{-(k+1)})$, the diagonal kills every consistent r.e. theory, dominated convergence, and the posterior on deterministic machines.

Truth and complexity. (a) By Lemma A.9(ii) with $\Lambda \equiv (x \neq x)$, every $\mathrm{Tick}_{k,j}$ is false, and by (iii) $\mathsf{Q} \vdash \mathrm{Tick}_{k,j} \leftrightarrow \sigma_{\Upsilon_0}(\underline{\langle k, j\rangle})$, since the second disjunct is Q-refutable: a Σ_1 sentence. So $\mathrm{Lot}^{\mathrm{ar}}_k$ is Q-equivalent to a false Σ_1 sentence and $\neg\mathrm{Tick}_{k,j^*_k}$ to a true Π^*_1 sentence. (b) By Lemma A.9(ii), $\mathrm{Tick}_{k,j}$ is true iff $j = j^\circ_k$: for $z = \ulcorner\Upsilon_1\urcorner$, $\mathrm{Ref}(z, k, j)$ holds iff $B \cup \mathsf{W}_k \vdash \neg\mathrm{Tick}_{k,j}$, and $\Lambda(z, \underline{\langle k, j\rangle})$ says exactly that j is the least refuted ticket, or 0 if none is refuted. Hence $\mathrm{Lot}^{\mathrm{ar}}_k$ is true, and Tick_{k,j^*_k} is false because $j^*_k \neq j^\circ_k$; every datum is true. By (iii), $\mathrm{Tick}_{k,j}$ is Q-equivalent to $\sigma_{\Upsilon_1}(\underline{\langle k, j\rangle}) \vee (\pi_{\Upsilon_1} \wedge \Lambda(\ulcorner\Upsilon_1\urcorner, \underline{\langle k, j\rangle}))$. With a numeral argument, (O) and (Σ_1C) unfold the bounded quantifiers of Λ and decide its Δ_0 parts, provably in Q ($t \le N_k$ is Δ_0 because the graph of 2^x is); so $\Lambda(\ulcorner\Upsilon_1\urcorner, \underline{\langle k, j\rangle})$ is Q-equivalent to a Boolean combination of the Σ_1 sentences $\mathrm{Ref}(\ulcorner\Upsilon_1\urcorner, \underline{k}, \underline{i})$, $i \le N_k$. Hence every datum is Q-equivalent to a Boolean combination of Σ_1 sentences. □

Proof of Corollary 1.5. (i) Theorem 1.4(b) with Theorem 1.1(a),(c). (ii) If $\mathbb{U}\{\omega \in E_\infty : B \cup A_\omega \text{ true}\} > 0$, Proposition A.4 gives a true r.e. theory proving the stream, contrary to Theorem 1.1(b). (iii) Fix k with $J := J_k \neq j^\circ_k$. The random lottery theory proves $\mathrm{Tick}_{k,J}$, which is Q-equivalent to $\sigma_{k,J} \vee (\pi_{\Upsilon_1} \wedge \Lambda_{k,J})$ with $\sigma_{k,J} := \sigma_{\Upsilon_1}(\underline{\langle k, J\rangle})$ a false Σ_1 sentence and $\Lambda_{k,J} := \Lambda(\ulcorner\Upsilon_1\urcorner, \underline{\langle k, J\rangle})$. Write Ref_i for $\mathrm{Ref}(\ulcorner\Upsilon_1\urcorner, \underline{k}, \underline{i})$. If some ticket is refuted and $J > j^\circ_k$, then $\mathrm{Ref}_{j^\circ_k}$ is a true Σ_1 sentence, so $\mathsf{Q} \vdash \mathrm{Ref}_{j^\circ_k}$ and $\mathsf{Q} \vdash \neg\Lambda_{k,J}$, and the theory proves the false Σ_1 sentence $\sigma_{k,J}$. If $J < j^\circ_k$, or no ticket is refuted (then $j^\circ_k = 0 < J$), then Q proves $\Lambda_{k,J} \to \mathrm{Ref}_J$ (for $J = 0 < j^\circ_k$ through $\mathsf{Q} \vdash \mathrm{Ref}_{j^\circ_k}$, which refutes the "none refuted" disjunct), and Ref_J is false; so the theory proves $\sigma_{k,J} \vee \mathrm{Ref}_J$, a false Σ_1 sentence. Given survival ($J_k \neq j^*_k$ for all k),

the events $J_k = j_k^\circ$ are independent with probabilities $1/N_k$, so all of them hold with probability $\prod_k 1/N_k = 0$. □

A computation (code in the accompanying repository) checks the bookkeeping of Theorem 1.4(b) and the three cases of (iii) exhaustively in a propositional model with up to eight tickets per lottery, for every behaviour of W_k and every winner.

Question A.10. On the stream of Theorem 1.4(b), is $P_\infty(\{\Sigma_1\text{-sound theories}\}) = 0$?

Corollary 1.5(ii) proves that the limit posterior lives on false theories, and (iii) that the random lottery theories among them are Σ_1-unsound; Proposition A.4's majority argument does not transfer to Σ_1-soundness, which is not a constraint of arity one.

A.4 Π_1 data

Throughout this subsection $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and B is decidable and true with $B \supseteq \mathsf{Q}$. Inc_0 is left-c.e. and not computable (**??**(a)), hence irrational. Put $I'_k := \lfloor 2^{k+2}\mathrm{Inc}_0 \rfloor 2^{-(k+2)}$, so that $I'_k \le \mathrm{Inc}_0 < I'_k + 2^{-(k+2)}$; given k, I'_k is given by $k+2$ bits. If D is of Π_1 type, then ψ_D is B-equivalent to $\pi_D \wedge \sigma_D$ with $\pi_D \in \Pi_1^*$ (the class is closed under conjunction) and σ_D a conjunction of true Σ_1 sentences; $B \vdash \sigma_D$ by (Σ_1C); so $B \vdash \psi_D \leftrightarrow \pi_D$, $O_{\psi_D} = O_{\pi_D}$ and $M_A(D) = \mathrm{Bel}_0(\pi_D)$. The definition is invariant under B-equivalence of literals, as it must be, since M_A, W_A and K_A depend on D only through $\mathrm{Cn}(D^{\pm})$ (Old).

Theorem A.11 (Threshold theories)**.** *There is c such that for every $k \ge 0$ the set*

$$T_k := \{\pi \in \Pi_1^* : \mathbb{U}(O_\pi) > I'_k + 2^{-(k+2)}\}$$

(i) is r.e., with an index e_k of complexity $K(e_k) \le k + K(k) + c$; (ii) contains every $\pi \in \Pi_1^*$ with $\mathrm{Bel}_0(\pi) \ge 2^{-k}$; (iii) consists of true sentences of positive prior belief, so $B \cup T_k$ is true, consistent and Σ_1-sound.

Proof. (i) $\mathbb{U}(O_\pi)$ is left-c.e. uniformly in π, and Π_1^* is decidable, so T_k is r.e. uniformly in (k, I'_k); a program consists of a self-delimiting description of k, the $k+2$ bits of I'_k, and a fixed code. (ii) $\mathbb{U}(O_\pi) = \mathrm{Inc}_0 + \mathrm{Bel}_0(\pi) \ge I'_k + 2^{-k} > I'_k + 2^{-(k+2)}$. (iii) If $\mathbb{U}(O_\pi) > I'_k + 2^{-(k+2)} > \mathrm{Inc}_0$, then $\mathrm{Bel}_0(\pi) > 0$: some tape has $B \cup A_\omega$ consistent and proving π, and π is true by (Π_1T). □

Proof of Theorem 1.6. π is emitted on the tape r iff $\mathbb{U}(O_\pi) > 0.r_{1:s} + 2^{-s}$ for some s; as $0.r_{1:s} + 2^{-s}$ decreases strictly to r for non-dyadic r, this holds iff $\mathbb{U}(O_\pi) > r$. If $r < \mathrm{Inc}_0$, every $\pi \in \Pi_1^*$ has $\mathbb{U}(O_\pi) \ge \mathrm{Inc}_0 > r$, including false ones such as $\neg(0=0)$, so the theory is inconsistent. If $r > \mathrm{Inc}_0$, every emitted π has $\mathrm{Bel}_0(\pi) > 0$ and is true by (Π_1T); B is true; so $B \cup A_r^{G_{\mathrm{thr}}}$ is true. For D of Π_1 type: if $\mathrm{Inc}_0 < r < \mathbb{U}(O_{\pi_D})$, the theory is consistent and contains π_D, and $B \vdash \psi_D \leftrightarrow \pi_D$; the dyadic tapes form a null set, so $\nu_{G_{\mathrm{thr}}}(D) \ge \mathbb{U}(O_{\pi_D}) - \mathrm{Inc}_0 = \mathrm{Bel}_0(\pi_D) = M_A(D)$. The upper bound is (Dom), with $K(\nu_{G_{\mathrm{thr}}}) \le K(G_{\mathrm{thr}}) + c$. □

A computation (code in the accompanying repository) checks Theorems 1.6 and A.11 in finite models of the prior.

Proof of Corollary 1.7. The first two inequalities are (Chain). For the last, let $k := \lceil L_A(D) \rceil$, so that $\mathrm{Bel}_0(\pi_D) = M_A(D) \ge 2^{-k}$. By Theorem A.11, $\pi_D \in T_k$, and $B \cup T_k$ is consistent and proves π_D, hence ψ_D. So $K_A(D) \le K(e_k) + c \le k + K(k) + c$. □

Remark A.12 (The coin-free version sums over levels). For every D of Π_1 type, $-\log W_A(D) \le L_A(D) + K^{\mathrm{pad}}(\lceil L_A(D)\rceil + 1) + c$. This is a bound on $-\log W_A$; the proved bound on K_A is the weaker $L_A + K(\lceil L_A\rceil) + c'$ of Corollary 1.7.

Proof. For a pair (j, L) with $L \ge 0$ and $0 \le j \le 2^L$ let $Z_{j,L} := \{\pi \in \Pi_1^* : \mathbb{U}(O_\pi) > j2^{-L}\}$. It is r.e., with an index $e(j, L)$ computable from the *pair* and injective in the pair (write the pair into the index), and $K(e(j, L)) \le K(j, L) + c \le L + K(L) + c'$, since j has $L + 1$ bits given L. If $j2^{-L} \in (\mathrm{Inc}_0, \mathbb{U}(O_{\pi_D}))$, then every member of $Z_{j,L}$ has $\mathrm{Bel}_0 > 0$ and is true by $(\Pi_1\mathrm{T})$, so $B \cup Z_{j,L}$ is true, contains π_D and proves ψ_D. With $k := \lceil L_A(D)\rceil$ the open interval $(\mathrm{Inc}_0, \mathbb{U}(O_{\pi_D}))$ has length $M_A(D) \ge 2^{-k}$, so for each $L \ge k + 1$ it contains at least $2^{L-k} - 1 \ge 2^{L-k-1}$ multiples of 2^{-L}. The indices $e(j, L)$ are distinct for distinct pairs, so

$$W_A(D) \;\ge\; \sum_{L \ge k+1} 2^{L-k-1}\, 2^{-L-K(L)-c'} \;=\; 2^{-k-1-c'} \sum_{L \ge k+1} 2^{-K(L)} \;\ge\; 2^{-k-1-c'-K^{\mathrm{pad}}(k+1)}.$$

Indexing the sets by the rational $j2^{-L}$ instead of by the pair would count $j2^{-L} = 2j2^{-(L+1)}$ once at every finer level, and the displayed sum would not bound W_A as written; a computation (code in the accompanying repository) confirms that the pair-indexed count meets the bound on all test intervals. □

Question A.13. On data of Π_1 type, is $K_A \le L_A + O(1)$?

The deterministic threshold needs the level written in full; the random threshold of Theorem 1.6 does not.

Corollary A.14 (Bounded loss on a Π_1 stream is one true theory)**.** *If every finite part of D_∞ is of Π_1 type and $\sup_n L_A(D_n) \le k$, then one true theory, $B \cup T_k$, of complexity at most $k + K(k) + c$, proves every datum; so $\sup_n -\log W_A(D_n) \le k + K(k) + c$.*

Proof. For each n, $\mathrm{Bel}_0(\pi_{D_n}) = M_A(D_n) \ge 2^{-k}$, so $\pi_{D_n} \in T_k$ by Theorem A.11(ii); T_k does not depend on n. By (iii), $B \cup T_k$ is true, and it proves every ψ_{D_n}; its index has complexity at most $k + K(k) + c$ by (i), which bounds $\sup_n -\log W_A(D_n)$ through **??**. □

Corollary A.15 (Chaitin's theorem for random theories)**.** *For all m and x, in the semimeasure convention: (a) if $K(x) \le m$, then $L_A(\{(\mathrm{Kgt}_m(x), 1)\}) = \infty$; if $K(x) > m$, then $L_A(\{(\mathrm{Kgt}_m(x), 1)\}) \ge m - K(m) - K(\lceil L_A\rceil) - c \ge m - 2\log m - O(\log\log m)$; (b) if $K(x) > m$, then $L_A(\{(\mathrm{Kgt}_m(x), 1)\}) \le K_A(\{(\mathrm{Kgt}_m(x), 1)\}) + c \le m + K(m) + c'$, and one true theory, B plus a single Π_1 sentence of complexity at most $m + K(m) + c$, proves $\mathrm{Kgt}_m(x)$ for every such x at once. The same bounds, with other constants, hold in the three-valued convention, in which the cost of the label 1 on this single datum is $-\log \mathrm{Bel}_\emptyset(\mathrm{Kgt}_m(x)) = L_A(\{(\mathrm{Kgt}_m(x), 1)\}) + \log(1 - \mathrm{Inc}_0)$.*

Proof. The deterministic bound (Chaitin). If $B \cup \mathsf{W}_e$ is consistent and proves $\mathrm{Kgt}_m(x)$, the program "given e and m, enumerate the theorems of $B \cup \mathsf{W}_e$ and output the first z such that $\mathrm{Kgt}_m(z)$ is among them" halts with some z, and $K(z) \le K(e) + K(m) + c$; $\mathrm{Kgt}_m(z)$ is true by $(\Pi_1\mathrm{T})$, so $K(z) > m$, whence $K(e) > m - K(m) - c$. So $K_A(\{(\mathrm{Kgt}_m(x), 1)\}) \ge m - K(m) - c$.

(b) Let s_m be the largest code of a halting computation of V on a program of length at most m (0 if there is none); it exists because there are finitely many such programs and each halting computation has one code (Uniq). Let $\mathrm{Bd}_m := \forall t\, \forall p\, \forall y\, (p \le \underline{\mathrm{rk}_m} \wedge \mathrm{Comp}(p, t, y) \to t \le \underline{s_m})$, a true Π_1 sentence. For x with $K(x) > m$, the Δ_0 sentence $\forall p \le \underline{\mathrm{rk}_m}\, \forall t \le \underline{s_m}\, \neg\mathrm{Comp}(p, t, \underline{x})$ is true, hence Q-provable, and together with Bd_m it gives $\mathrm{Kgt}_m(x)$ by pure logic. s_m is computable from

m and the number $H_m \le 2^m$ of halting programs of length at most m (Kraft's inequality), an $(m+1)$-bit number: run the programs until H_m of them have halted. So $K(\mathrm{Bd}_m) \le K(m)+m+c$. Since $B \cup \{\mathrm{Bd}_m\}$ is true, hence consistent, and proves $\mathrm{Kgt}_m(x)$, **??**(c) gives $K_A \le K_{\mathrm{fin}} + c \le m + K(m) + c'$; and $L_A \le K_A + c$ by (Chain). Without unique codes a halting program could have infinitely many codes, and Bd_m would be false.

(a) For $K(x) \le m$ the datum is a false Π_1 sentence, and $M_A = 0$ by (Π_1T). Otherwise Corollary 1.7 and the deterministic bound give $L_A \ge K_A - K(\lceil L_A \rceil) - c' \ge m - K(m) - K(\lceil L_A \rceil) - c''$. By (b), $\lceil L_A \rceil \le m + K(m) + c$, so $K(\lceil L_A \rceil) \le \log m + O(\log\log m)$, and $K(m) \le \log m + O(\log\log m)$. For the three-valued convention: for a single two-valued datum $M^3_A(\{(\varphi,1)\}) = M_A(\{(\varphi,1)\})$, and $M^3_A(\emptyset) = 1 - \mathrm{Inc}_0$, so the three-valued cost of the label 1 is $-\log \mathrm{Bel}_\emptyset(\mathrm{Kgt}_m(x)) = L_A(\{(\mathrm{Kgt}_m(x),1)\}) + \log(1-\mathrm{Inc}_0)$, which differs from L_A by a constant. □

The deterministic step is Chaitin's bound (Chaitin, 1974), used only in its upper-bound form (a consistent extension of Q that proves $\mathrm{Kgt}_m(x)$ has complexity greater than $m - K(m) - c$); it makes no claim about least constants, which depend on the coding of machines (Raatikainen, 1998). One m-bit Π_1 law explains every true complexity lower bound at level m. Yet the datum carries at most one bit of label information: unconstrained SOI predicts it at cost $O(1)$ by always answering "true", and the lower bound of **??** is $O(1)$ here. In the normalised convention the m bits disappear:

Proposition A.16 (The normalised convention: deciding, not predicting)**.** *Let B be decidable and true with $\mathsf{PA} \subseteq \mathrm{Cn}(B)$. There is c such that for all m, d and $j \ge 0$, all strings $x \in \{0,1\}^d$ except at most $2^{2m+K(m)+j+c}+2^{m+1}$ of them satisfy $K(x) > m$ and $\mathrm{Dis}_\emptyset(\mathrm{Kgt}_m(x)) \le 2^{-j}\mathrm{Bel}_\emptyset(\mathrm{Kgt}_m(x))$; so the loss of $(\mathrm{Kgt}_m(x),1)$ in the normalised convention is at most $\log(1+2^{-j}) \le 1.45 \cdot 2^{-j}$.*

Proof. Step 1: few refutable strings per theory. Let $x_1, \dots, x_{2^{m+1}}$ be distinct strings. By (O), $\mathsf{Q} \vdash p \le \underline{\mathrm{rk}_m} \to \bigvee_{q \le \mathrm{rk}_m} p = \underline{q}$, so $\neg\mathrm{Kgt}_m(x_i)$ is Q-equivalent to $\bigvee_{q\le \mathrm{rk}_m} \exists t\, \mathrm{Comp}(\underline{q}, t, \underline{x_i})$. Hence $\bigwedge_{i \le 2^{m+1}} \neg\mathrm{Kgt}_m(x_i)$ is Q-equivalent to the disjunction, over all maps $f : \{1,\dots,2^{m+1}\} \to \{0,\dots,\mathrm{rk}_m\}$, of $\bigwedge_i \exists t\, \mathrm{Comp}(\underline{f(i)}, t, \underline{x_i})$. There are $\mathrm{rk}_m + 1 = 2^{m+1} - 1$ programs, fewer than the strings, so every f has $f(i) = f(i')$ for some $i \ne i'$, and then (Uniq) gives $x_i = x_{i'}$, which Q refutes. So $\mathsf{PA} \vdash \neg \bigwedge_{i \le 2^{m+1}} \neg\mathrm{Kgt}_m(x_i)$: every consistent extension of B refutes $\mathrm{Kgt}_m(x)$ for at most $2^{m+1}-1$ strings x. Integrating over the consistent tapes (Tonelli), $\sum_x \mathrm{Bel}_0(\neg\mathrm{Kgt}_m(x)) \le 2^{m+1}$.

Step 2: a lower bound on belief. For $K(x) > m$, Corollary A.15(b) and (Dom) give $\mathrm{Bel}_0(\mathrm{Kgt}_m(x)) \ge 2^{-m-K(m)-c}$.

Step 3: Markov's inequality. By Step 1, fewer than $2^{m+1} \cdot 2^{j+m+K(m)+c}$ strings have $\mathrm{Bel}_0(\neg\mathrm{Kgt}_m(x)) > 2^{-j}2^{-m-K(m)-c}$, and fewer than 2^{m+1} strings have $K(x) \le m$. For every other x, Step 2 gives $\mathrm{Bel}_0(\neg\mathrm{Kgt}_m(x)) \le 2^{-j}\mathrm{Bel}_0(\mathrm{Kgt}_m(x))$; and $\mathrm{Dis}_\emptyset$, $\mathrm{Bel}_\emptyset$ are $\mathrm{Bel}_0(\neg\cdot)$, $\mathrm{Bel}_0(\cdot)$ divided by the same $1-\mathrm{Inc}_0$. The normalised probability of the label 1 is $\mathrm{Bel}_\emptyset/(\mathrm{Bel}_\emptyset+\mathrm{Dis}_\emptyset) \ge 1/(1+2^{-j})$, and $\log(1+y) \le y/\ln 2$. □

The m bits are the price of deciding the sentence, not of predicting its truth value: every Π_1 affirmation must be backed by a true theory, while a consistent extension of PA refutes $\mathrm{Kgt}_m(x)$ for fewer than 2^{m+1} strings. So for $d \ge 2m + K(m) + j + c'$ almost every string of length d costs essentially nothing in the normalised convention, while every x with $K(x) > m$ costs $m \pm O(\log m)$ bits in the semimeasure and three-valued conventions (Corollary A.15). In the three-valued convention most of the remaining mass goes to "independent": by Step 1, $\sum_x \mathrm{Dis}_\emptyset(\mathrm{Kgt}_m(x)) \le 2^{m+1}/(1-\mathrm{Inc}_0)$, so for most x the inducer predicts "independent" with probability $1 - 2^{-m+O(\log m)}$. One-tape consistent SOI behaves the same way: **??** transfers

$M_A(\{(\varphi,b)\})$ to $M_F^{\mathrm{cons}}(\{(\varphi,b)\})$ up to factors $2^{\pm c}$, which leaves the semimeasure statement unchanged and shifts j by a constant in the normalised one. With $B=\mathsf{Q}$ alone, Step 1 needs Q to prove the uniqueness of outputs, which it may not; the normalised behaviour is then open. A computation (code in the accompanying repository) checks the finite bookkeeping of Steps 1 and 3 in finite models in which uniqueness of outputs is imposed.

Proposition A.17 (True Π_1^*-axiomatised theories)**.** *Let T be an r.e. set of true Π_1^* sentences and $k:=\lceil -\log \mathbb{U}_A(\mathrm{Ext}T)\rceil$. Then*

$$-\log \mathbb{U}_A(\mathrm{Ext}T)-c \;\le\; -\log \mathbf{m}(\mathrm{Ext}T) \;\le\; -\log \mathbb{U}_A(\mathrm{Ext}T)+K(k)+c.$$

Proof. Right: each $\pi\in T$ has $\mathrm{Bel}_0(\pi)\ge \mathbb{U}_A(\mathrm{Ext}T)\ge 2^{-k}$, since every consistent theory extending $\mathrm{Cn}(T)$ proves π; so $T\subseteq T_k$. $B\cup T_k$ is consistent, so $e_k\in \mathrm{Ext}(T)$ and $\mathbf{m}(\mathrm{Ext}T)\ge \mathbf{m}(e_k)\ge 2^{-k-K(k)-c}$. Left: $\mathbb{U}_A(\mathrm{Ext}T)\ge 2^{-c}\mathbf{m}(\mathrm{Ext}T)$ (**??**; $B\cup T$ is consistent because it is true). □

The same comparison for the exact class $[T]$, and for general T, is part of **??**.

Proposition A.18 (One false Σ_1 datum)**.** *Let $\mathcal{L}=\mathcal{L}_{\mathrm{ar}}$, B decidable and true with $B\supseteq\mathsf{Q}$, and Υ_0 the formula of Theorem 1.4(a). For every $d\ge 2$ there are distinct $x_1,x_2\in\{0,1\}^d$ such that $D:=\{(\Upsilon_0(\underline{x_1}),0),(\Upsilon_0(\underline{x_2}),1)\}$ (strings coded by length-lexicographic rank) has $K_A(D)>\log d-2$ and $L_A(D)\le 2+c$. The literal $\neg\Upsilon_0(\underline{x_1})$ is Q-equivalent to a true Π_1^* sentence, and $\Upsilon_0(\underline{x_2})$ to a false Σ_1 sentence.*

Proof. Upper bound. The machine that, for every n, reads a fresh bit b_n and emits $\Upsilon_0(\underline{n})^{b_n}$ has a consistent theory on every tape (Lemma A.9(i) and compactness) and proves D with probability $\frac14$; apply (Dom). *Lower bound* (the colouring argument of **??**). Let $\ell:=\lceil \log(d/\log 3)\rceil-1$, so that $2^\ell<d/\log 3$ and $3^{2^\ell-1}<2^d$. Fewer than 2^ℓ programs have length less than ℓ. Colour each $x\in\{0,1\}^d$ by the vector, over these programs p, of "+" if $V(p)$ halts with an index e such that $B\cup\mathsf{W}_e$ is consistent and proves $\Upsilon_0(\underline{x})$, "−" if it is consistent and proves $\neg\Upsilon_0(\underline{x})$, and "∗" otherwise. Two distinct x_1,x_2 share a colour, and then no program shorter than ℓ gives a consistent theory proving $\neg\Upsilon_0(\underline{x_1})\wedge\Upsilon_0(\underline{x_2})$. So $K_A(D)\ge\ell\ge\log d-\log\log 3-1>\log d-2$. The complexity claims are Theorem 1.4(a)'s. □

Proposition A.19 (Two true data)**.** *Let $\mathcal{L}=\mathcal{L}_{\mathrm{ar}}$, and let B be decidable and true with $B\supseteq\mathsf{Q}$. There is a formula Υ_2 such that for every $d\ge 2$ some distinct $x_1,x_2\in\{0,1\}^d$ give true data $D_d:=\{(\Upsilon_2(\underline{x_1}),0),(\Upsilon_2(\underline{x_2}),1)\}$ with $K_A(D_d)>\log d-2$ and $L_A(D_d)\le 2+c$. Each datum is Q-equivalent to a Boolean combination of Σ_1 sentences (hence, by prenexing, both to a Σ_2 and to a Π_2 sentence).*

Proof. The colour formulas. For a program p (a numeral) put

- $\mathrm{Halt}_p :\equiv \exists t\,\exists e\,\mathrm{Comp}(\underline{p},t,e)$, a Σ_1 sentence ($V(p)$ halts);
- $\mathrm{Ok}_p :\equiv \forall t\,\forall e\,\big(\mathrm{Comp}(\underline{p},t,e)\to \mathrm{Con}(B\cup\mathsf{W}_e)\big)$, a Π_1 sentence, with $\mathrm{Con}(B\cup\mathsf{W}_e)$ the canonical Π_1 formula in e;
- $\mathrm{Prv}_p^b(z,y) :\equiv \exists t\,\exists e\,\big(\mathrm{Comp}(\underline{p},t,e)\wedge \mathrm{Prov}_{B\cup\mathsf{W}_e}(\ulcorner\varphi_{[z]}(\dot y)^b\urcorner)\big)$, a Σ_1 formula, for $b\in\{0,1\}$.

The colour entry of y at p is "+" iff $\mathrm{Halt}_p\wedge\mathrm{Ok}_p\wedge\mathrm{Prv}_p^1(z,y)$, "−" iff $\mathrm{Halt}_p\wedge\mathrm{Ok}_p\wedge\mathrm{Prv}_p^0(z,y)$, and "∗" otherwise. In $\mathbb{N}$, where $V(p)$ has at most one output, these say exactly what the colouring in the proof of Proposition A.18 says (with $\varphi_{[z]}$ for Υ_0). The consistency condition is written with a universal quantifier because "$\exists e\,(\mathrm{Comp}(p,t,e)\wedge\mathrm{Con}(\dots))$" would be Σ_2. Let $\Lambda(z,x)$ say: "x is

the code of a string of some length $d \geq 2$, and x is the second element of the lexicographically first pair of distinct strings of length d that have the same colour", colours taken over the programs of length less than $\ell_d := \lceil \log(d/\log 3) \rceil - 1$, with all quantifiers outside the colour formulas bounded by x. Let Υ_2 be Lemma A.9's formula for Λ, and (x_1, x_2) that pair for $z = \ulcorner \Upsilon_2 \urcorner$; it exists by the pigeonhole count of Proposition A.18. By Lemma A.9(ii), $\Upsilon_2(\underline{x_2})$ is true and $\Upsilon_2(\underline{x_1})$ is false, so both data are true. The fair-coin machine of Proposition A.18, with Υ_2 in place of Υ_0, gives $L_A(D_d) \leq 2 + c$ by Lemma A.9(i). The pair has the same colour for the actual Υ_2, which gives $K_A(D_d) \geq \ell_d > \log d - 2$ as in Proposition A.18. For a numeral argument, (O) and (Σ_1C) unfold the bounded quantifiers of $\Lambda(\ulcorner \Upsilon_2 \urcorner, \underline{x})$ and decide its Δ_0 parts in Q, leaving a Boolean combination of the Σ_1 sentences Halt_p, Prv_p^b and the Π_1 sentences Ok_p; with Lemma A.9(iii), each datum is Q-equivalent to a Boolean combination of Σ_1 sentences, hence, by prenexing, both to a Σ_2 and to a Π_2 sentence. □

A.5 Gödel phenomena

Proof of Proposition 1.8. (a) The posterior lives on consistent extensions of Q, and all of them prove σ by (Σ_1C). (b) Some conjunct of π is false; its negation is a true Σ_1 sentence, so every consistent extension of Q proves $\neg\pi$. (c) Consider the r.e. theories $D^{\pm} \cup \{\pi\}$, $D^{\pm} \cup \{\neg\pi\}$ and $D^{\pm}$. The first is true in a model of $B \cup D^{\pm}$ whose $\mathcal{L}_{\mathrm{ar}}$-reduct is $\mathbb{N}$, so it is consistent with B, and it proves π. The second is consistent with B because $B \cup D^{\pm} \nvdash \pi$ (by the deduction theorem for the finite set $D^{\pm}$), and it proves $\neg\pi$. The third proves neither: not π by hypothesis, and not $\neg\pi$ because $B \cup D^{\pm}$ is true and $\neg\pi$ is false. All three prove ψ_D and have positive weight by (Dom). (d) Apply (c) to the Π_1^* sentence $\neg\sigma$. □

An extreme case. If the data present $T := \mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$, a consistent and Σ_1-unsound theory, every consistent extension of T proves $\neg\mathrm{Con}(T)$, since $\mathsf{PA} \vdash \mathrm{Con}(T) \to \mathrm{Con}(\mathsf{PA})$; so $\mathrm{Dis}(\mathrm{Con}(T)) = 1$ although $\mathrm{Con}(T)$ is true. A computation (code in the accompanying repository) checks (a)–(d) and the one-sided error pattern on random instances of a finite model in which (Σ_1C) is the only arithmetical input.

Proof of Theorem 1.9. By the s-m-n theorem there is an injective computable g with $\mathsf{W}_{g(e)} = \mathsf{W}_e \cup \{\neg\mathrm{Con}(B \cup \mathsf{W}_e)\}$, the canonical consistency statement of the presentation e. If $B \cup \mathsf{W}_e$ is consistent, (G2) gives $B \cup \mathsf{W}_e \nvdash \mathrm{Con}(B \cup \mathsf{W}_e)$, so $B \cup \mathsf{W}_{g(e)}$ is consistent; its new axiom is a false Σ_1 sentence; and it proves everything W_e proves. As $\mathbf{m}(g(e)) \geq 2^{-c}\mathbf{m}(e)$ and $K(g(e)) \leq K(e) + c$, summing over the Σ_1-sound compatible e, whose images are distinct, gives the claim, and the posterior bound follows from $W_A(\mathcal{U}; D) \geq 2^{-c} W_A(\overline{\mathcal{U}}; D)$. A true theory is Σ_1-sound, which gives "a fortiori". □

The event that $B \cup A_\omega$ is Σ_1-unsound is $\bigcup_\sigma O_\sigma$ over the false Σ_1 sentences σ, an open set, so $\mathbb{U}_A(\mathcal{U}; D)$ and $P_D(\mathcal{U})$ are defined.

Proposition A.20 (One-tape soundness is controlled by the coding gap)**.** *Under the hypotheses of Theorem 1.9, for every consistent D,*

$$P_D(\mathcal{U}) \;\geq\; 2^{L_A(D) - K_A(D) - c}.$$

Proof. Let e witness $K_A(D)$. Its companion $g(e)$ from the proof of Theorem 1.9 is consistent, Σ_1-unsound, proves ψ_D, and has $K(g(e)) \leq K_A(D) + c$; by (Dom) the tapes that begin with a shortest program for it have measure at least $2^{-K_A(D)-c}$, and their theory is $\mathsf{W}_{g(e)}$. Divide by $M_A(D) = 2^{-L_A(D)}$. □

So one-tape SAI can become confident of soundness only on data where the coding theorem fails badly, which is where randomness dominates (Section 1.3). On data of Π_1 type, Proposition A.20 and Corollary 1.7 give $P_D(\mathcal{U}) \geq 2^{-K(\lceil L_A(D)\rceil)-c}$, a bound that degrades as the loss grows; the threshold machine removes the degradation.

Theorem A.21 (Soundness is never learned from Π_1 data)**.** *Let B be decidable and true with $\mathsf{PA} \subseteq \mathrm{Cn}(B)$. There is c such that for every D of Π_1 type with $M_A(D) > 0$, $\mathbb{U}_A(\mathcal{U}; D) \geq 2^{-c} M_A(D)$; so $P_D(\overline{\mathcal{U}}) \leq 1 - 2^{-c}$.*

Proof. For a finite string w let U_w be the set of sentences that G_{thr} (Theorem 1.6) emits on some tape extending w; it is r.e. uniformly in w, and $U_w = \{\pi \in \Pi_1^* : \mathbb{U}(O_\pi) > 0.w\}$, since the emissions under prefixes $v \sqsupseteq w$ have thresholds $0.v + 2^{-|v|}$ with infimum $0.w$, not attained, and those under shorter prefixes $u \sqsubseteq w$ have thresholds at least $0.w + 2^{-|w|}$. Hence U_w is true if $0.w > \mathrm{Inc}_0$, and is all of Π_1^*, an inconsistent set, if $0.w < \mathrm{Inc}_0$ (equality is impossible, since Inc_0 is irrational). Moreover $U_{w'} \subseteq U_w$ for $w \sqsubseteq w'$, provably in PA from the definitions (every prefix of w' is comparable with w). Let $G^{\mathrm{u}}_{\mathrm{thr}}$ emit everything G_{thr} emits and, for every prefix w of the tape, $\chi_w := \neg\mathrm{Con}(B \cup U_w)$.

Consistency. Let $r > \mathrm{Inc}_0$ be a non-dyadic tape and F a finite subset of $A_r^{G^{\mathrm{u}}_{\mathrm{thr}}}$, with companion sentences $\chi_{w_1}, \ldots, \chi_{w_j}$; let w^* be the longest w_i. If $0.w^* < \mathrm{Inc}_0$, every $U_{w_i} \supseteq U_{w^*}$ is inconsistent, every χ_{w_i} is a true Σ_1 sentence, provable from B, and the rest of F lies in the true theory $A_r^{G_{\mathrm{thr}}}$. If $0.w^* > \mathrm{Inc}_0$, then $B \cup U_{w^*}$ is consistent, r.e. and contains PA, so $B \cup U_{w^*} \cup \{\chi_{w^*}\}$ is consistent by (G2); it contains $F \cap A_r^{G_{\mathrm{thr}}}$, because $A_r^{G_{\mathrm{thr}}} = \{\pi : \mathbb{U}(O_\pi) > r\} \subseteq U_{w^*}$ as $r > 0.w^*$; and it proves each χ_{w_i} with $w_i \sqsubseteq w^*$, because $U_{w^*} \subseteq U_{w_i}$ provably, so a proof of $\bot$ from $B \cup U_{w^*}$ is one from $B \cup U_{w_i}$.

Unsoundness. $r > \mathrm{Inc}_0$ has a prefix w with $0.w > \mathrm{Inc}_0$, since the truncations increase to r; for it χ_w is a false Σ_1 sentence.

Measure. For $\mathrm{Inc}_0 < r < \mathbb{U}(O_{\pi_D})$ the theory of $G^{\mathrm{u}}_{\mathrm{thr}}$ is consistent, Σ_1-unsound and contains π_D, and $B \vdash \psi_D \leftrightarrow \pi_D$. So $\nu_{G^{\mathrm{u}}_{\mathrm{thr}}}$ gives the event "consistent, unsound and proving ψ_D" probability at least $M_A(D)$, as in the proof of Theorem 1.6, and (Dom), tape by tape, gives $\mathbb{U}_A(\mathcal{U}; D) \geq 2^{-K(G^{\mathrm{u}}_{\mathrm{thr}})-c} M_A(D)$. □

A computation (code in the accompanying repository) checks the dyadic and prefix facts of this proof in finite models with non-dyadic weights.

What is known about Question 1.10. With B true and $\mathsf{PA} \subseteq \mathrm{Cn}(B)$, the answer is yes for data of Π_1 type (Theorem A.21) and for data with a bounded coding gap $K_A - L_A$ (Proposition A.20). On the Win-lottery stream with $\mathcal{L} = \mathcal{L}_{\mathrm{ar}} \cup \{\mathrm{Win}\}$ it is yes as well: the machine "G_{lot} plus the axiom $\neg\mathrm{Con}(B)$" has the same survival event; its theories are consistent (a model of $B + \neg\mathrm{Con}(B)$ exists by (G2), has distinct numerals as every model of Q does, and is expanded by the winners) and Σ_1-unsound; so $P_{D_n}(\mathcal{U}) \geq 2^{-c} \prod_k (1 - 2^{-(k+1)})$ for every n. On the arithmetic lotteries the random lottery theories are themselves unsound (Corollary 1.5(iii) for (b); in (a) they affirm the false Σ_1 sentences Tick_{k,J_k}). In general we can prove only the following. For a theory machine G let U_w^G be the set of sentences that G emits on some tape extending w, and $\mathcal{K}^G := \{\omega : B \cup U_w^G \text{ is consistent for some prefix } w \text{ of } \omega\}$. Under the hypotheses of Theorem 1.9, for every D,

$$\mathbb{U}_A(\mathcal{U}; D) \;\geq\; 2^{-K(G)-c}\, \mathbb{U}(\{\omega : B \cup A_\omega^G \text{ consistent}, \vdash \psi_D\} \cap \mathcal{K}^G).$$

Indeed, the companion G^{u} emits what G emits and $\neg\mathrm{Con}(B\cup U^G_w)$ for every prefix w of the tape, and $K(G^{\mathrm{u}})\le K(G)+c$; on a tape ω on which $B\cup A^G_\omega$ is consistent, the longest-prefix argument above shows that $B\cup A^{G^{\mathrm{u}}}_\omega$ is consistent ($A^G_\omega\subseteq U^G_{w^*}$ because w^* is a prefix of ω), and on $\mathcal{K}^G$ some added sentence is a false Σ_1 sentence; then apply (Dom) tape by tape. This answers the question for no class of data by itself: it fails for coin machines, all of whose sets U^G_w are inconsistent. The difficulty in general is that a companion must add a false Σ_1 sentence that the random theory A_ω does not refute, chosen from finitely much of the tape. $\neg\mathrm{Con}(A_\omega)$ is not a sentence, and A_ω may prove $\mathrm{Con}(Z)$ for every r.e. Z read off a finite prefix (a machine can emit, at each stage, the consistency of everything it has emitted so far). The natural companions need either a consistent U_w, which fails for coin machines, or a fixed false Σ_1 sentence that the random theory leaves open, which fails for such reflective machines.

Proposition A.22 ($\mathrm{Con}(B\cup T)$ given data from T)**.** *Let T be r.e., with $B\cup T$ true and $\mathsf{PA}\subseteq\mathrm{Cn}(B\cup T)$, and let the data present T: $\mathrm{Cn}(D^{\pm}_\infty)=\mathrm{Cn}(B\cup T)$. Write χ_T for $\mathrm{Con}(B\cup T)$ or the Gödel sentence of $B\cup T$. (a) $\lim_n\mathrm{Bel}_{D_n}(\chi_T)$, $\lim_n\mathrm{Dis}_{D_n}(\chi_T)$ and $\lim_n\mathrm{Indep}_{D_n}(\chi_T)$ exist and lie in $(0,1)$. (b) The belief and independent shares certify the truth of χ_T in the sense of Proposition 1.8(b). (c) Which of "true" and "false" gets more mass is a choice of universal machine: for every universal prefix machine V and every j there are universal prefix machines V', V'' under which the ratio of the limits of $\mathrm{Bel}(\chi_T)$ and $\mathrm{Dis}(\chi_T)$ is at least 2^j-1, respectively at most $1/(2^j-1)$.*

Proof. (a) By (Lim) (**??**(d)), the limit posterior is the prior restricted to the consistent extensions of $B\cup T$, which is incomplete by (G2). Three deterministic theories have positive weight. $T+\mathrm{Con}(B\cup T)$ is true, because $B\cup T$ is true, hence consistent, and $\mathrm{Con}(B\cup T)$ is a true Π_1 sentence. $T+\neg\mathrm{Con}(B\cup T)$ is consistent with B by (G2) for the presentation $B\cup T\supseteq\mathsf{PA}$. T decides neither: not $\mathrm{Con}(B\cup T)$ by (G2), and not its negation because $B\cup T$ is true and $\neg\mathrm{Con}(B\cup T)$ is false. Each of the three limits is therefore positive, and each is less than 1 because the other two are positive. (b) is Proposition 1.8(b) applied to the Π_1 sentence $\mathrm{Con}(B\cup T)$. (c) Let $V'(0^i1)$ be an index of the deterministic machine for $T+\mathrm{Con}(B\cup T)$ for $i<j$, and $V'(0^jp):=V(p)$; V' is a universal prefix machine. The tapes that begin with some 0^i1, $i<j$, have measure $1-2^{-j}$ and give a compatible theory proving $\mathrm{Con}(B\cup T)$; all tapes on which $\neg\mathrm{Con}(B\cup T)$ is proved begin with 0^j. So, within E_∞, the masses of the two events are at least $1-2^{-j}$ and at most 2^{-j}; dividing both by the common normaliser $\mathbb{U}(E_\infty)$ gives the ratio. V'' is symmetric, with $T+\neg\mathrm{Con}(B\cup T)$. The Gödel sentence of $B\cup T$ is PA-provably equivalent to $\mathrm{Con}(B\cup T)$ (Smoryński, 1977), so the same holds for it. □

The truth hypothesis is needed. With $B:=\mathsf{PA}+\neg\mathrm{Con}(\mathsf{PA})$, which is decidable and consistent by (G2), and $T:=\emptyset$: since $\mathsf{PA}\vdash\mathrm{Con}(B)\to\mathrm{Con}(\mathsf{PA})$, $B\vdash\neg\mathrm{Con}(B)$, so every compatible theory refutes $\mathrm{Con}(B\cup T)$, $\lim\mathrm{Dis}=1$, and (a) and (c) fail. A computation (code in the accompanying repository) checks the construction of (c) in a finite model.

Proposition A.23 (Old evidence)**.** *If $B\vdash\varphi^b$ for every (φ,b) in D, then $M_A(D)=M_A(\emptyset)$ and the posterior equals the prior; likewise for W_A and K_A. In particular, with $\mathsf{Q}\subseteq\mathrm{Cn}(B)$ and a Δ_0 formula $P(x)$, the data $\{(P(\underline{n}),1):n\le N\}$ (true, since otherwise they are inconsistent with B) leave $\mathrm{Bel}(\forall x\,P(x))$ at its prior value, for every N.*

Proof. If $B\vdash\varphi^b$ for every $(\varphi,b)\in D$, then $\mathrm{Cn}(D^{\pm})=\mathrm{Cn}(\emptyset)$, so $M_A(D)=M_A(\emptyset)$, $E_D=E_\emptyset$, and the same for W_A and K_A (Old). With $\mathsf{Q}\subseteq\mathrm{Cn}(B)$ and P a Δ_0 formula, every true instance $P(\underline{n})$ is Q-provable by (Σ_1C), and a false one is Q-refutable, so its datum would be inconsistent with B. □

Instances of a fresh predicate (empirical induction, $B = \emptyset$) do confirm $\forall x\, R(x)$, up to the ω-gap (Proposition 1.11). So the inference from checked instances needs a reasoner for whom computing $P(\underline{n})$ is costly: logical induction (Garrabrant et al., 2016a) is one, and a derivation-charged axiom inducer, as in Section 6 of our earlier report (Hänni and Claude (Anthropic), 2026), is another. Whether a derivation-charged posterior then concentrates on provers of $\forall x\, P(x)$ depends on competitors that prove the instances cheaply without proving the universal (compare Proposition A.27); we do not settle this.

A.6 The ω-gap

Proof of Proposition 1.11. I is a consistent r.e. theory proving the stream, so $\mathbb{U}(E_\infty) \geq 2^{-K(I)-c} > 0$ (**??**(c)) and (Lim) applies; by **??**(d) with $\Gamma = I$, P_∞ is the prior restricted to the consistent extensions of I, whether or not I is complete. In (i) and (ii) every consistent extension of I proves what I proves. In (iii), I, $I + \forall x\, \varphi$ and $I + \neg\forall x\, \varphi$ are consistent r.e. theories with positive weight, which prove $\forall x\, \varphi$, refute it, and decide it neither way, respectively; and a consistent extension of I refutes $\forall x\, \varphi$ iff it is ω-inconsistent at φ, since it proves every instance. □

Two standard cases. With $B = \emptyset$ and the atomic diagram of a computable structure this is **??**. With $B \supseteq \mathsf{Q}$ and $\varphi \in \Delta_0$ the instances are free (Proposition A.23) and the posterior does not move at all; case (iii) holds iff B decides $\forall x\, \varphi$ neither way. For Σ_1-sound B this is iff $B \nvdash \forall x\, \varphi$: with all instances true, $\neg\forall x\, \varphi$ is logically equivalent to the false Σ_1 sentence $\exists x\, \neg\varphi$, which a Σ_1-sound B does not prove. Without Σ_1-soundness this fails: for $B = \mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$ and $\varphi(x)$ the Δ_0 formula "x is not a PA-proof of $\bot$", every instance is true and Q-provable, so $\mathrm{Cn}(I) = \mathrm{Cn}(B)$, and $B \vdash \neg\forall x\, \varphi$: case (ii) holds although $B \nvdash \forall x\, \varphi$.

Proposition A.24 (The Gaifman condition)**.** *Under the hypotheses of Proposition 1.11, enumerate* $\mathrm{CT} = \{t_0, t_1, \dots\}$. *Then* $\lim_n \mathrm{Bel}_0(\bigwedge_{i\leq n} \varphi(t_i))$ *exists, and*

$$\lim_n \mathrm{Bel}_0\Big(\bigwedge_{i\leq n} \varphi(t_i)\Big) - \mathrm{Bel}_0(\forall x\, \varphi) = \mathbb{U}\{\omega : B \cup A_\omega \text{ consistent, proves every } \varphi(t_i),\ \nvdash \forall x\, \varphi\}.$$

This is 0 *iff* $I \vdash \forall x\, \varphi$, *and otherwise at least* $2^{-K(I)-c}$, *where* $K(I)$ *is the complexity of an index of* I. *The same holds for every posterior* Bel_D *with* D *proved by* I, *after dividing by* $M_A(D)$.

Proof. The events $\{\omega : B \cup A_\omega$ consistent and proves $\varphi(t_0), \dots, \varphi(t_n)\}$ decrease in n to the event of proving every instance; continuity from above gives the limit. A theory proving $\forall x\, \varphi$ proves every instance, so the difference is the stated measure. If $I \vdash \forall x\, \varphi$, every consistent theory proving all instances proves $\forall x\, \varphi$, and the difference is 0. Otherwise I itself lies in the event, with weight at least $2^{-K(I)-c}$ by (Dom). For Bel_D, divide by $M_A(D)$. □

"Prove", not "decide", is the right word: in the ω-inconsistent case (ii) of Proposition 1.11 the instances refute the universal, yet the gap equals the whole consistent mass of the extensions of I.

Theorem A.25 (A uniform Gödelian ω-gap)**.** *Let* B *be decidable and true with* $\mathsf{PA} \subseteq \mathrm{Cn}(B)$. *For every* k *there is a* Δ_0 *formula* $\zeta_k(x)$ *such that* $\forall x\, \zeta_k$ *is true, every instance* $\zeta_k(\underline{n})$ *is a true* Δ_0 *sentence (so* Q*-provable, and instance data are free by Proposition A.23), and* $2^{-k-K(k)-c} \leq \mathrm{Bel}_0(\forall x\, \zeta_k) < 2^{-k}$. *Hence after any instance data the posterior belief is* $\mathrm{Bel}_0(\forall x\, \zeta_k)/(1 - \mathrm{Inc}_0) < 2^{-k+c'}$, *and* $\mathrm{Dis} + \mathrm{Indep} > 1 - 2^{-k+c'}$.

Proof. Let $\forall x\,\zeta_k$ be the canonical consistency statement $\mathrm{Con}(B \cup T_k)$ of the presentation e_k (Theorem A.11), rewritten with one universal quantifier over a code of a pair (a derivation of $\bot$, an enumeration stage), with membership in B and in T_k certified inside the Δ_0 matrix by computation witnesses and the quantifiers contracted by Δ_0 pairing. For such a natural rewriting $\mathsf{PA} \vdash \forall x\,\zeta_k \leftrightarrow \mathrm{Con}(B \cup T_k)$ (standard). $B \cup T_k$ is true (Theorem A.11(iii)), so $\forall x\,\zeta_k$ is true and its instances are true Δ_0 sentences. If $\mathrm{Bel}_0(\forall x\,\zeta_k) \geq 2^{-k}$, Theorem A.11(ii) puts $\forall x\,\zeta_k$ into T_k, so $B \cup T_k \vdash \forall x\,\zeta_k$, hence, by the PA-provable equivalence, $B \cup T_k \vdash \mathrm{Con}(B \cup T_k)$, contradicting (G2) for the consistent r.e. presentation $B \cup T_k \supseteq \mathsf{PA}$. The lower bound is (Dom) for the true one-sentence theory $\{\forall x\,\zeta_k\}$, whose index is computable from e_k, of complexity at most $K(e_k) + c \leq k + K(k) + c'$. The posterior after B-entailed data is the prior normalised by $M_A(\emptyset) = 1 - \mathrm{Inc}_0$, that is $\widehat{\mathrm{Bel}_0} = \mathrm{Bel}_\emptyset$, so the posterior belief is $\mathrm{Bel}_0(\forall x\,\zeta_k)/(1-\mathrm{Inc}_0) < 2^{-k+c'}$. □

Remark A.26 (Where the gap closes). (a) Under the ω-rule, with $B \supseteq \mathsf{Q}$, every random theory that is consistent under that rule has $\mathcal{L}_{\mathrm{ar}}$-content $\mathrm{Th}(\mathbb{N})$: the inducer is omniscient about arithmetic a priori. (b) Under full second-order consequence the loss on any presentation of $\mathrm{Th}(\mathbb{N})$ is at most $K(\mathsf{PA}^2) + c$: the unlearnability of $\mathrm{Th}(\mathbb{N})$ (**??**) rests on effective consequence. (c) The generative variant closes the gap only with the universal sentence among the data: by Theorem 3.9(c1) of the earlier report, with a faithful likelihood and $\forall x\,\varphi$ in the data space, the posterior probability of "proves $\forall x\,\varphi$" tends to $\mathbf{1}[T^* \vdash_{\mathsf{C}} \forall x\,\varphi]$, whether the generator T^* proves it, not to its truth value; for filtered instance data the gap stays open (its part (d1)).

The closures of Remark A.26. (a) *The ω-rule.* Let $\vdash_\omega$ be closure under first-order deduction and the ω-rule (from $\varphi(\underline{n})$ for all n, infer $\forall x\,\varphi$). Then $\mathsf{Q} \vdash_\omega \chi$ for every true $\mathcal{L}_{\mathrm{ar}}$-sentence χ: write χ in prenex form with a Δ_0 matrix and argue by induction on the prefix. True Δ_0 sentences are Q-provable; a true $\exists x\,\psi(x)$ has a true instance $\psi(\underline{n})$, provable by induction, and $\exists$-introduction applies; a true $\forall x\,\psi(x)$ has all instances true and provable, and the ω-rule applies. So with $B \supseteq \mathsf{Q}$ every $\vdash_\omega$-consistent random theory has $\mathcal{L}_{\mathrm{ar}}$-content exactly $\mathrm{Th}(\mathbb{N})$: it proves every truth and, being consistent, no falsehood. The gap closes because the inducer is omniscient about arithmetic a priori. (b) *Full second-order consequence* (given Dedekind's categoricity theorem, standard). Let hypotheses be second-order sentences, and let compatibility mean "has a full model, and every full model satisfies ψ_D". Every full model of second-order Peano arithmetic PA^2 is isomorphic to $\mathbb{N}$, so $\{\mathsf{PA}^2\}$ is compatible with every finite correctly labelled data set. By the dominance argument (the tapes that begin with a program for $\{\mathsf{PA}^2\}$), the loss on any presentation of $\mathrm{Th}(\mathbb{N})$, in any order, is at most $K(\mathsf{PA}^2) + c$, and the limit posterior has $\mathrm{Bel}(\varphi) = \mathbf{1}[\mathbb{N} \models \varphi]$ for every $\mathcal{L}_{\mathrm{ar}}$-sentence, trivially, since every sentence is a datum; the content is the loss bound. This places the unlearnability of $\mathrm{Th}(\mathbb{N})$ (**??**) on effective consequence, not on the complexity of arithmetical truth. (c) *The generative variant.* Theorem 3.9 of the earlier report concerns the generative inducer over a countable class with i.i.d. data from a generator T^*. Its parts (a), (b): the posterior probability of "proves $\forall x\,\varphi$" tends to the prior share of provers in the class of theories with the generator's law. Its part (c1): in the faithful regime, with $\forall x\,\varphi$ in the data space, the limit is $\mathbf{1}[T^* \vdash_{\mathsf{C}} \forall x\,\varphi]$, whether the generator proves the universal in the calculus C used there, not the truth value of $\forall x\,\varphi$. So this closes the gap only in the sense that the universal sentence is itself observed. For filtered data the gap stays open: for quantifier-free φ in the minimal class of that paper, the limit is 0 under its plain likelihood and the prior share under its selection-aware likelihood (its part (d1)), and with an unknown filter it is a filter-prior mixture (its part (e)). The generative variant is therefore not a closure of the ω-gap on instance data.

Line budgets. Let hypotheses be single sentences ψ, and call a datum χ *ℓ-compatible* with ψ if $B \cup \{\psi\}$ is consistent and proves χ in a fixed Hilbert calculus by a derivation of at most ℓ lines (lines, not symbols). The data are the closed instances $\varphi(S^n 0)$, $n \in \mathbb{N}$.

Proposition A.27 (Line budgets and a Kreisel-type property)**.** *Assume that some sentence is ℓ-compatible with every datum (for instance $\forall x\, \varphi$, when $B \cup \{\forall x\, \varphi\}$ is consistent and ℓ is at least the constant number of lines of instantiation). Then for any prior with full support on sentences, the limit posterior gives* $\mathrm{Bel}(\forall x\, \varphi) = 1$ *iff every ψ that is consistent with B and proves every $\varphi(S^n 0)$ within ℓ lines also proves $\forall x\, \varphi$ (a* Kreisel property *of (B, φ, ℓ)).*

Proof. The set of sentences compatible with the first n data decreases in n to the set of ψ that are consistent with B and ℓ-compatible with every datum, which is nonempty by hypothesis. If every such ψ proves $\forall x\, \varphi$, the limit belief is 1; a ψ in it that does not prove $\forall x\, \varphi$ has positive weight, and then the limit belief is less than 1. □

With $B = \emptyset$ and $\varphi(x) := x = 0 \vee \exists y\, x = Sy$, the property already fails at a constant budget, but that example is degenerate: every instance is logically valid, so the data are old evidence (Proposition A.23), and the budget plays no role. The following example is not degenerate.

Example A.28 (A non-degenerate failure)**.** Let $B = \emptyset$, $\mathcal{L} = \{0, S, R\}$ with R a fresh unary predicate, and data $R(S^n 0)$; they are not valid, since the one-point structure with R empty refutes $R(0)$. With $\varphi(x) := x = 0 \vee \exists y\, x = Sy$, let $\psi_1 := \forall x\, (\varphi(x) \to R(x))$ and $\psi_2 := \forall x\, R(x)$. Each instance $\varphi(S^n 0)$ has a derivation of at most five lines (an equality axiom $S^n 0 = S^n 0$, the axiom $\chi(t) \to \exists y\, \chi(y)$ with $t = S^{n-1} 0$ and modus ponens, the axiom $\beta \to \alpha \vee \beta$ and modus ponens; three lines for $n = 0$), so ψ_1 derives $R(S^n 0)$ in nine lines (seven for $n = 0$): the hypothesis, its instance at $S^n 0$ by $\forall$-elimination and modus ponens, the derivation of $\varphi(S^n 0)$, and modus ponens. ψ_2 derives each datum in three lines. The structure $\{0, 1, 2, a\}$ with $S(0) = 1$, $S(1) = 2$, $S(2) = 2$, $S(a) = 0$ and $R := \{0, 1, 2\}$ satisfies ψ_1 (the only element outside R is a, and $\varphi(a)$ fails) and every $R(S^n 0)$ (closed terms denote 0, 1 or 2), but not $\forall x\, R(x)$. So with $\ell = 9$ both ψ_1 and ψ_2 stay ℓ-compatible with every datum, ψ_1 is consistent and does not prove $\forall x\, R(x)$, and under any full-support prior the limit belief in $\forall x\, R(x)$ lies strictly between 0 and 1: the Kreisel property fails for $(\emptyset, R, 9)$, with instances that are not valid.

A computation (code in the accompanying repository) checks the derivations line by line for $n \leq 60$ in a small Hilbert-style proof checker, and the countermodel.

Kreisel's conjecture says that if PA proves $\varphi(S^n 0)$ in at most k lines for every n, then $\mathsf{PA} \vdash \forall x\, \varphi$; it is open for the usual Hilbert-style PA, with partial results and counterexamples for nearby theories (Parikh, 1973; Baaz and Pudlák, 1993; Hrubeš, 2007, 2009; Santos and Kahle, 2021). These concern infinite theories with schemata. The property of Proposition A.27 concerns finite extensions $B \cup \{\psi\}$, in which a k-line derivation can use only the finitely many axioms of ψ and logical axioms; we know of no result on it, so the link is a reformulation.

Question A.29. For $B = \mathsf{Q}$ and Δ_0 formulas φ, does every finite ψ that proves all instances of φ in boundedly many lines prove $\forall x\, \varphi$?

Over Q with $\varphi \in \Delta_0$ the instances are old evidence again; the question is which finite theories the line-budget inducer keeps, not what it learns from the instances.

A.7 True arithmetic

Let $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$, $B \subseteq \mathrm{Th}(\mathbb{N})$ decidable, $(\varphi_i)_{i\geq 1}$ a computable enumeration of all sentences, D_n the first n of them labelled by truth, and $b_{1:n}$ the labels. The loss is unbounded and at most $n + K(n) + c$ (**??**). It grows at least polynomially, because with binary numerals the sentences "the k-th digit of Ω_V is 1" occur early.

Theorem A.30 (A polynomial lower bound)**.** *Suppose the enumeration lists sentences by length, and within a length in a fixed order, and that (1) at most $C\kappa^L$ sentences have length at most L, for every L, and (2) there are terms $\underline{k}^{\mathrm{b}}$ with value k and length at most $c_1 \log k + c_2$. Put $\xi := 1/(c_1 \log \kappa)$ and $m(n) := \lfloor (n/C')^{\xi} \rfloor$, where C' depends only on the syntax and on a fixed formula for the binary digits of Ω_V. Then for all n, in the semimeasure convention: (a) $K(b_{1:n} \mid n) \geq m(n) - K(n) - c$; (b) $L_A(D_n) \geq C'^{-\xi} n^{\xi} - K(n) - c$; (c) $L_A(D_n) \geq (2C')^{-\xi} n^{\xi} - 2\log\log n - c$ for $n \geq 4$.*

Proof. Let $\Omega_s \uparrow \Omega_V$ be the computable approximations. Since Ω_V is irrational, its k-th binary digit is the eventual value of the k-th digit of Ω_s. Let $\mathrm{Bit}'(y)$ be a formula with $\mathbb{N} \models \mathrm{Bit}'(\underline{k})$ iff that digit is 1: "there is s such that for all $s' \geq s$ the y-th digit of $\Omega_{s'}$ is 1", with the digit predicate expressed by a Δ_0 formula about a computation witness. Put $\mathrm{Bit}(x) := \exists y\, (y = x \wedge \mathrm{Bit}'(y))$, so that x occurs once, and $\mathrm{Bit}_k := \mathrm{Bit}(\underline{k}^{\mathrm{b}})$. Then $|\mathrm{Bit}_k| \leq |\mathrm{Bit}| + c_1 \log k + c_2$, and the position of Bit_k in the enumeration is at most $C\kappa^{|\mathrm{Bit}|+c_2}\kappa^{c_1 \log k} = C' k^{1/\xi}$. The first n sentences therefore contain $\mathrm{Bit}_1, \ldots, \mathrm{Bit}_{m(n)}$, at positions computable from k, and their labels are the digits $\Omega_{1:m(n)}$. So $\Omega_{1:m(n)}$ is computable from n and $b_{1:n}$, and $K(\Omega_{1:m(n)}) \leq K(b_{1:n} \mid n) + K(n) + c$. By Chaitin's theorem $K(\Omega_{1:m}) \geq m - c$. This is (a). (b) is (a) with **??**(b) and $m(n) \geq (n/C')^{\xi} - 1$. (c) $L_A(D_n)$ is nondecreasing in n, because $E_{D_{n+1}} \subseteq E_{D_n}$. For $n \geq 4$ let $n' := 2^{\lfloor \log n \rfloor} \in (n/2, n]$; then $K(n') \leq K(\lfloor \log n \rfloor) + c \leq 2\log\log n + c'$, and by monotonicity and (b), $L_A(D_n) \geq L_A(D_{n'}) \geq (n/(2C'))^{\xi} - 2\log\log n - c''$. □

The exponent for a stated syntax. Take the Polish-notation syntax with fourteen symbols: terms 0, St, $+tt$, $\cdot tt$ and variables $x' \cdots'$ (x followed by $i \geq 0$ primes); formulas $=tt$, $<tt$, $\neg f$, $\wedge ff$, $\vee ff$, $\to ff$, $\forall vf$, $\exists vf$; and binary numerals $\underline{1}^{\mathrm{b}} = S0$, $\underline{2m}^{\mathrm{b}} = \cdot\, \underline{m}^{\mathrm{b}}\, SS0$, $\underline{2m+1}^{\mathrm{b}} = S{\cdot}\, \underline{m}^{\mathrm{b}}\, SS0$, so that $c_1 = 5$ and $c_2 = 2$. Counting all strings ($\kappa = 14$) gives $\xi \geq 1/(5\log 14) > 0.0525$. Counting formulas gives more. With $V(x) = x/(1-x)$, the generating function of the variables, the generating functions of terms and formulas satisfy

$$T = x + V + xT + 2xT^2, \qquad F = 2xT^2 + xF + 3xF^2 + 2xVF,$$

and every term on the right carries a factor x. If rationals $x_0, T^*, F^* > 0$ satisfy both equations with "$\geq$" in place of "=", then the iterates of the right-hand sides from 0 have nonnegative coefficients, agree with the series up to degree k after k steps, and stay below (T^*, F^*) at x_0; so $\sum_L F_L x_0^L \leq F^*$, and the number of formulas, hence of sentences, of length at most L is at most $\frac{F^*}{1-x_0} x_0^{-L}$ for every L. A computation with exact rational arithmetic (code in the accompanying repository) verifies such a super-solution at $x_0 = 1000/5385$, with $T^* \approx 0.800042$ and $F^* \approx 0.609287$; this gives $C = 0.7482$ and $\kappa = 5.385$, hence $\xi \geq 1/(5\log 5.385) > 0.0823$. The same computation finds no certificate at $\kappa = 5.38$; the growth rate of the formula counts is about 5.383.

Only the numeral length and the growth of the syntax enter ξ; the (large) size of Bit enters only the constant. With unary numerals the position of Bit_k is at most $C''\kappa^k$, so $m(n)$ is only about $\log_\kappa n$; then (b) is vacuous as written, since $K(n)$ can be about $\log n > \log_\kappa n$, but (c)'s monotonicity argument still gives $L_A(D_n) \geq \log_\kappa n - 2\log\log n - c$. The bound is weak in absolute

terms (for $n = 10^6$, $n^{0.0823}$ is about 3.1); it is a statement about the shape of the growth, not a usable estimate.

The digit sentences among the first n carry about $m(n)$ incompressible bits (Chaitin, 1975), and a true theory decides correctly, so:

Corollary A.31 (Quantitative incompleteness)**.** *Let T be r.e. with $B \cup T \subseteq \mathrm{Th}(\mathbb{N})$, and let $u_n(T)$ be the number of $i \le n$ such that $B \cup T$ proves neither φ_i nor $\neg\varphi_i$. Then $u_n(T) \ge C'^{-\xi} n^{\xi}/(2 \log n)$ for all large n.*

Proof. $b_{1:n}$ is computable from an index e_T of T, n, the set of undecided positions among the first n and their labels, since a true theory decides correctly. Using $\log \binom{n}{u} \le u \log(en/u)$ and $K(u \mid n) \le \log n + c$, this gives $K(b_{1:n} \mid n) \le K(e_T) + u_n(T)(\log n + 3) + 2 \log n + c$. Compare with Theorem A.30(a), which is the inequality needed here (the statement (b) about L_A would not suffice): $u_n(T)(\log n + 3) \ge (n/C')^{\xi} - K(n) - K(e_T) - 2 \log n - c'$, and the right-hand side grows like n^{ξ} while the subtracted terms are $O(\log n)$. □

Conversely, the loss is small if some true r.e. theory leaves few of the first n sentences undecided:

Proposition A.32 (An upper bound by a reduction)**.** *For every r.e. T with $B \cup T \subseteq \mathrm{Th}(\mathbb{N})$ and every n,*

$$L_A(D_n) \le K(e_T) + \log \tbinom{n}{u_n(T)} + u_n(T) + 2 \log n + O(\log \log n).$$

Hence if some such T has $u_n(T)/n \to 0$, then $L_A(D_n) = o(n)$.

Proof. $B \cup T$ together with the correctly labelled undecided sentences among the first n is a consistent (true) r.e. theory proving D_n, with an index computable from e_T, n, u_n, the positions ($\log \binom{n}{u_n}$ bits, given n and u_n) and the u_n labels; then (Chain) gives $L_A \le K_A + c$. If $u_n/n \to 0$, then $\log \binom{n}{u_n} \le nH(u_n/n) = o(n)$, with H the binary entropy function. □

Question A.33. In such an enumeration of the sentences of $\mathcal{L}_{\mathrm{ar}}$ labelled by truth, is $L_A(D_n) = o(n)$?

By Proposition A.32, a positive answer to Question A.33 would follow if some true r.e. theory decided a set of sentences of asymptotic density one in the enumeration, that is, if the truth of arithmetic were generically decidable in the sense of generic-case complexity (Myasnikov and Rybalov, 2008). Such questions depend on the representation: the halting problem is decidable on a set of asymptotic probability one for one model of Turing machines (Hamkins and Miasnikov, 2006). Calude and Jürgensen (2004) show that the probability that a true sentence of length n is provable tends to 0, in a probability space of statements with n-bit codes under their numbering. If a version of this held for the length enumeration used here, every true r.e. theory would leave a positive fraction of the sentences of each length undecided, and the sufficient condition of Proposition A.32 would fail for every T. It would not settle Question A.33: $K(b_{1:n} \mid n)$ could still be $o(n)$, since truth values along the enumeration are highly redundant (for instance ψ and $\neg\neg\psi$). Corollary A.31 gives only a polynomial lower bound on u_n.

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